Crowding

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Abstract

Crowding is a common disamenity, yet little is known about its monetary cost. We estimate the willingness to pay (WTP) to avoid crowding in public transit using administrative smartcard trip records from the Beijing Subway and a revealed-preference approach. Passengers choose departure times, trading off fares, in-train crowding, and schedule deviation, with fare variation from an early-bird discount. Leveraging the subway's network structure, we impute real-time in-train crowding, infer each passenger's optimal arrival time, and construct a novel instrumental variable for crowding. The marginal WTP to reduce crowding by one passenger per square meter is about 0.05 RMB per minute, implying a crowding externality that exceeds the fare. High-income passengers have higher WTP to avoid crowding but are less price-sensitive. An optimal crowding tax improves welfare within the subway system but is regressive; a two-class configuration elicits self-selection and benefits both high- and low-income commuters.

Keywords: Crowding externality, Public transit, Travel demand.
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1 Introduction

Many goods decline in quality as more people use them at the same time. Consider a grocery store before a major holiday or a beach on the first hot weekend of summer. As more people arrive,

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individuals have less room and feel increasingly uncomfortable. This is crowding, defined as sharing a limited space with many others. Each additional person makes the space more crowded and increases the discomfort experienced by others, imposing a negative externality. Quantifying this externality requires knowing how much people are willing to pay to avoid crowding. We estimate this parameter in the Beijing Subway during the morning peak, a setting where crowding is a common feature of daily urban life: for people living in densely populated areas, commuting by public transit is the most common, and often the most severe, crowding experience of the day.

Crowding on public transit has a broad impact on riders. Dissatisfaction with crowding is associated with greater stress and lower life satisfaction (Kumagai et al., 2021), and improvements in commuting conditions, including reduced crowding, can improve worker performance (Lu et al., 2024). Crowding also shapes how transit services are provided. In addition to expanding capacity through more frequent services or larger vehicles, operators can use fares to encourage travel outside peak hours, offer different service classes, or even limit passenger entry through queuing when crowding becomes severe.[1] The willingness to pay (WTP) to avoid crowding is a key input for evaluating these policies, yet credible estimates remain limited.

There are at least three empirical challenges. First, measurement: high-frequency data on the crowding individuals experience are rare at scale. Crowdedness changes as people enter and leave, so measuring exposure requires knowing how densely occupied a space is and how long each person spends in it; most existing studies therefore rely on stated-preference surveys rather than observed choices. Second, other attributes move with crowding: speed, service, and status. Buses are more crowded when roads are congested and travel is slower; the less crowded option, business class on an airplane or a skybox in a stadium, comes bundled with service quality and status that are hard to measure. Observed choices therefore do not reveal the value of crowding alone. Third, endogeneity arising from correlated demand shocks: places are crowded because many people want them at the same time. Work hours, an exciting game, or a holiday sale draw people in while also making the trip, the game, or the shopping more valuable, so crowding is endogenous and may be positively associated with demand even when people dislike it.

We address these challenges using administrative subway ride records in Beijing in 2016, focusing on the morning peak. These records report each passenger's entry and exit stations and times but not where they were in between. We therefore develop an algorithm that combines these records with the subway network and train timetables to recover passengers' real-time locations and, from there, measure crowding exposure for each ride.[2] Moreover, the Beijing Subway is a well-suited

[1]For example, London's Piccadilly line upgrade includes plans for more frequent peak service. Hong Kong's MTR offers an early-bird fare discount to encourage passengers to travel earlier. In the Tokyo region, JR East offers premium Green Cars alongside ordinary cars on commuter trains, with more spacious seating available for an additional fare. In Shenzhen, busy metro stations restrict entry during peak hours, requiring passengers to queue before entering.

[2]The algorithm jointly estimates each passenger's fastest route and the average access, transfer, and exit times at each station, with in-train time taken from published train timetables. Dividing the number of passengers sharing the same route segment in the same time bin by the available train floor area yields crowding density in persons per square meter.

setting for isolating crowding disutility from the confounders common in transit. Subway cars are identical, and trains run with near-perfect punctuality, so travel time is nearly constant conditional on the route, varying little with crowding. To tackle the endogeneity of crowding, we construct an instrument for crowding using passengers whose trips overlap incidentally with the commuters we study because of the network structure but reflect different travel demands.

Using these novel, high-frequency measures, we characterize crowding patterns during the morning peak in Beijing. Among trips originating from the 32 selected stations in our main analysis, the average density is about 3.4 persons per square meter and the ninetieth percentile reaches roughly 5 persons per square meter. The average in-train exposure lasts about 38 minutes. We show empirically that crowding does not meaningfully slow down trains or induce detour routing, supporting the interpretation of the in-train crowding experience as nearly pure discomfort. Crowding also remains stable across dates conditional on the route and departure time. Commuters therefore face a predictable crowding profile across the morning, which justifies modeling their choices as responses to expected crowding, measured by the mean for the same route, day of week, and time bin.

To estimate how much commuters are willing to pay to avoid this crowding, we model how they choose when to depart, taking each commuter's origin–destination (OD) pair and route as given. Operationally, we divide departure times into 15-minute bins and estimate a discrete-choice model in which commuters trade off fares, expected crowdedness, and the cost of deviating from their optimal arrival time. Commuters can also choose an outside option consisting of other travel modes. The extent to which passengers pay a higher fare for less crowding or a smaller schedule deviation reveals their marginal willingness to pay (MWTP) to avoid crowding and rescheduling. The demand model includes OD-by-date and date-by-time-bin fixed effects, which absorb daily demand differences across routes and departure-time patterns common across routes on each date.[3] Crowding relief may be more valuable on more crowded trips and to some income groups than others. We examine this heterogeneity using a quadratic crowding specification and a random-coefficient model that allows the disutility from fares, crowding, and rescheduling to vary with income.

Estimating how commuters trade off fares, expected crowding, and schedule deviations poses three identification problems: fares and crowding each require variation unrelated to demand shocks, and optimal arrival times are unobserved. To identify the fare response, we exploit exogenous variation in fares across departure times generated by an off-peak discount. The discount, known as the Early-Bird Discount (EBD), gives a 30 percent fare reduction to passengers who tap in at one of 16 stations on two suburban lines before 7 AM on workdays. Control stations without the discount help identify common departure-time patterns, allowing us to separate the fare response at EBD sta-

Integrating this density over the duration of a passenger's trip gives their total in-train crowding exposure, measured in person-minutes per square meter.

[3]These demand fixed effects allow daily conditions, such as weather, to affect demand differently across OD pairs and allow the common morning demand profile to vary by date.

tions from these patterns. The EBD stations are among the busiest suburban stations rather than a random draw; identification uses the fare step at 7 AM within an OD pair on a date, so station-level differences are absorbed by fixed effects.

To identify crowding disutility, we use an instrument that addresses the correlation between crowding and unobserved demand. The instrument is based on incidental companion passengers (ICPs), who travel from other origins to distant destinations but share train segments with a focal passenger, adding to crowding on those segments. Unobserved demand shocks change the utility of traveling on a particular OD pair, date, and departure-time bin, beyond the common rush-hour patterns absorbed by the fixed effects. We restrict ICPs to infrequent riders and assume that, conditional on the model's fixed effects, their demand shocks are unrelated to those of the focal market, following a shared-capacity identification strategy similar to that of Hortaçsu et al. (2024) in the airline market. We define incidental companion time (ICT) as the weighted sum of ICPs, where each ICP is weighted by the duration of shared travel. The instrument is expected ICT, the expectation over dates for the same OD pair, day of week, and time bin, constructed exactly as expected crowding. We supplement expected ICT with instruments that interact EBD-origin status with the discount cutoff and functions of departure time. Because the discount induces passengers to reschedule, it changes crowding even within periods when fares are unchanged. These instruments exploit variation in expected crowding that remains after controlling for fares and the model's other controls.

To measure schedule deviations, we need each passenger's optimal arrival time, a well-known challenge in the structural travel demand literature because it may vary both across passengers and across dates for the same passenger. We infer optimal arrival times from *coworkers*, defined as passengers who arrive at the same destination on the same day but live elsewhere and take routes not directly affected by the EBD. Their arrival distribution reflects arrival choices without the direct fare incentive to shift schedules. Within each destination-date cell, we reweight this distribution to match the income composition of EBD riders. We then rank EBD riders by their actual arrival time and assign each rider the coworker arrival time at the same percentile of the reweighted distribution. The method is non-parametric and uses only contemporaneous data, reducing concerns about changes in commuting patterns over time.

The baseline estimates imply substantial costs of crowding. The average passenger is willing to pay 0.053 RMB per minute to reduce crowding density by one person per square meter. Over the average in-train exposure of 38 minutes, this reduction is worth roughly 2.0 RMB, compared with an average fare of 5.3 RMB. At the average density documented above, total crowding disutility is about 6.7 RMB, or 16 percent of the median hourly wage of about 43 RMB. The MWTP rises with the level of crowding, so welfare costs are concentrated on the most congested trips: under a quadratic specification, a ride in a nearly empty car costs under 1 percent of the hourly wage, while a ride at the 95th percentile of density, 6 persons per square meter, costs about 23 percent. The baseline MWTP to avoid rescheduling is 0.144 RMB per minute.

Allowing preferences to vary with income reveals substantial differences in crowding valuations. In the random-coefficient model, higher-income riders dislike crowding more and are less sensitive to fares, so their MWTP for crowding reduction rises sharply with income. At the median monthly income of 7,500 RMB, passengers are willing to pay 0.07 RMB per minute to reduce crowding density by one person per square meter; at the 90th percentile of income, 17,500 RMB, this amount rises to 1.62 RMB. As a result, few high-income passengers take the subway in the equilibrium under the current fare structure. In the random-coefficient model, the MWTP to avoid rescheduling, by contrast, is approximately 0.10 RMB per minute and relatively flat across the income distribution. High-income and low-income passengers place similar value on arriving on time, even as they differ markedly in how much they are willing to pay to avoid a crowded car.

Building on these estimates, we investigate the welfare consequences of three alternative policies: a Pigouvian crowding tax, quantity control through queuing, and a revenue-neutral two-class configuration with higher-priced Business cars and discounted Standard cars. To keep computation manageable while retaining the key policy trade-offs, we model a representative trip and abstract from network interactions and scheduling dynamics. The optimal crowding tax is 3.30 RMB, about two-thirds of the baseline fare. The tax raises welfare within the subway system, but its incidence is sharply regressive. Low-income riders lose consumer surplus, while high-income riders gain slightly even before the revenue is rebated. Subway ridership declines by 16 percent, driven by a large reduction in low-income riders. Quantity control targets the same ridership reduction but delivers lower welfare because time wasted in the queue dissipates economic rent. Both policies push passengers out of the subway, generating additional welfare losses from road congestion. Including these spillovers, welfare under the crowding tax falls slightly below baseline.

The two-class configuration, by contrast, delivers a Pareto improvement. It exploits the steep income gradient in WTP to avoid crowding: passengers who value crowding relief more pay for less-crowded service. Under revenue-neutral pricing, the optimal Business fare is 10.30 RMB, roughly twice the baseline fare, while the Standard fare is 2.44 RMB, a discount of about 52 percent. High- and low-income passengers self-select into different classes, with Standard cars more than three times as crowded as Business cars. High-income passengers benefit substantially, primarily from less-crowded cars. Low-income passengers experience far more crowded cars, but they are compensated by deeply discounted fares, and their consumer surplus increases slightly on net. The discounted Standard fare draws new riders from the outside option, and total ridership rises by about 18 percent. Attracting more people to use the subway alleviates surface road congestion, especially when high-income residents switch from cars. Including these spillovers, total welfare rises by about 15 percent.

Contributions and Related literature. This paper contributes to three strands of literature. First, we contribute to the valuation of crowding in public transit. A growing empirical literature studies demand elasticity in public transit (e.g., Davis, 2021; Hahn et al., 2023) and evaluates off-peak discounts as a tool against crowding (Halvorsen et al., 2016; Anupriya et al., 2020; Gu et al., 2024),

but these studies do not estimate the disutility of crowding itself. We are aware of no study that estimates within-system crowding disutility in monetary terms from a transit system's own fare variation. We emphasize that the subway setting differs conceptually from the literature on road congestion (e.g., Vickrey, 1969; Arnott et al., 1993; Small and Verhoef, 2007; Anderson, 2014; Yang et al., 2020; Kreindler, 2024). Road congestion primarily generates externalities through lost time, whereas crowding in public transit generates disutility through physical discomfort even when travel times are unaffected (Tirachini et al., 2013; Wardman and Whelan, 2011; Haywood and Koning, 2015).

Much of the evidence on crowding valuation comes from passenger surveys that elicit stated preferences (e.g., Douglas and Karpouzis, 2006; Haywood and Koning, 2015; Singh et al., 2023), as reviewed by Li and Hensher (2011) and Wardman and Whelan (2011). Stated-preference estimates rest on hypothetical choices, which, as the contingent-valuation literature documents, need not match actual behavior and are sensitive to the framing and scope of the question (Diamond and Hausman, 1994; Hausman, 2012; Lu et al., 2008). The studies closest to ours estimate passenger preferences from observed choices in trip-level records. Hörcher et al. (2017) and Bansal et al. (2022) use route choices recorded by smart cards in Asian metro systems, Yap et al. (2020) Dutch tram and bus trips, and Tirachini et al. (2016) the willingness of Singapore riders to travel backward to secure a seat. These papers estimate crowding costs relative to travel time, so monetary valuation requires an external value of time. The one revealed-preference study that values crowding in money, Peer et al. (2016), measures crowding only by a coarse, fixed indicator of expected train occupancy and leaves its endogeneity unaddressed, which may explain why it finds no significant crowding cost. We instead build an economic model that incorporates fares, crowding, and schedule deviation jointly and estimate WTP to avoid crowding in monetary terms using exogenous fare variation generated by the EBD. We further address the endogeneity of crowding with a novel instrument and estimate preference heterogeneity by income through a random-coefficient specification.

Second, we contribute to the literature on optimal pricing under transit congestion externalities. Prior theoretical work characterizes transit pricing and subsidies with crowding (Parry and Small, 2009; de Palma et al., 2017; Coulombel and Monchambert, 2023). To our knowledge, we provide the first empirical estimate of an optimal crowding tax based on monetary crowding valuations identified from exogenous fare variation. We also compare the tax against quantity controls that ration access through waiting rather than price. The welfare comparison connects to the theory of price versus quantity controls (Weitzman, 1974) and empirical work on market and nonmarket allocation of automobile licenses (Li, 2018). Under heterogeneous disutility from crowding, we show that second-degree price discrimination through a two-class configuration elicits self-selection and produces a Pareto improvement, a result that complements Cook and Li (2025) on high-occupancy toll lanes.

Third, we develop new measurement tools that exploit the granular trip-level data and the network structure of the subway system. Although trip-level transit records are increasingly used in

empirical work (e.g., Gu et al., 2024), they typically contain only entry and exit times and stations and do not allow researchers to trace passengers through the network. We develop an iterative algorithm that imputes each passenger's optimal route, real-time location, and in-train crowding. We also propose a novel way of measuring the optimal arrival time, a crucial but typically unobserved input in a large class of structural travel models with endogenous congestion, including the bottleneck model (Vickrey, 1969; Arnott et al., 1993). Our approach uses the arrival distribution of riders unaffected by the EBD to infer EBD riders' optimal arrival times non-parametrically through quantile matching. We contribute to studies of scheduling flexibility among commuters (e.g., Gu et al., 2024; Kreindler, 2024; Hall, 2024). The same network structure also provides the incidental-companion instrument for crowding. Together, these tools allow us to measure crowding and rescheduling disutility and may prove useful in other settings with similarly granular data.

The rest of the paper is organized as follows. Section 2 describes the institutional setting and the data. Section 3 introduces the demand model and provides an overview of the estimation strategy. Section 4 describes the construction of key variables and instruments and presents summary statistics. Section 5 documents stylized facts that motivate the modeling choices. Section 6 presents the estimation results for both the homogeneous logit and the random-coefficient specifications. Section 7 conducts the welfare and counterfactual analysis. Section 8 concludes.

2 Background and Data

2.1 Background

Beijing is China's capital and second most populous city. In the first two decades of the 21st century, it has built one of the world's largest subway systems, almost from scratch. By 2016, the network consisted of 19 lines and more than 340 stations, carrying 3.7 billion annual passenger trips. The subway is especially important for long-distance commuting because of its speed and reliability. Excluding walking trips, the subway accounted for 15% of commuting trips and roughly 40% of commuting passenger mileage in 2014 (Beijing Transport Institute, 2015). A typical subway trip covers about 15 kilometers and takes approximately 37 minutes, including time spent accessing, waiting, and exiting.

The fare for a subway ride is a step function of the track distance between the origin and the destination. A single trip starts at 3 RMB for distances under 6 kilometers, with the fare increasing by 1 RMB for each additional increment of up to 20 kilometers, eventually reaching 10 RMB for trips between 92 and 112 kilometers. Starting in December 2015, an early-bird discount (EBD) was introduced at 16 stations on two suburban lines, the Batong Line and the Changping Line. Passengers who tap in at one of these stations before 7 AM receive a 30% fare reduction.

The 16 EBD stations are among the busiest during the morning rush, but they are not the only

busy stations. Other suburban lines connecting dense residential neighborhoods with the city center exhibit similar ridership patterns. The EBD was later expanded to eight additional stations on Line 6 by the end of 2016, which falls outside our sample period. We select 16 stations on Lines 5 and 6 as comparison stations. Line 6 runs parallel to the Batong Line, and the eight stations we select received the EBD after our sample period. Line 5 cuts through the city center from north to south, with its northern branch paralleling the Changping Line and passing through dense residential neighborhoods. The eight Line 5 stations we choose have been in active discussion for inclusion in the EBD program. Appendix Figure C.3 shows the location of EBD and control stations in the Beijing Subway network.

Figure 1 compares tap-in counts and in-train crowding density at EBD and control stations during morning hours. The two groups exhibit similar crowding profiles, with density on downtown-bound trains peaking at about 2.7 to 2.8 persons per square meter shortly after 8 AM. The similarity supports the use of control stations as a valid comparison group. The figure also reveals the behavioral response to the EBD. At EBD stations, tap-ins display a visible dent just after 7 AM, with a corresponding bunching just before the cutoff. This pattern is consistent with riders advancing their departures to qualify for the 30% discount. No such discontinuity appears at control stations, where tap-ins rise smoothly through the morning peak. The crowding density, which cumulates boardings from upstream stations, also shows a discontinuity around the cutoff, albeit less sharp than that in tap-ins. The discontinuities generated by the EBD may seem small and localized in the narrow neighborhood around the time cutoff, but as Figure 2 illustrates, as the EBD affects the crowdedness of subway cars, its impact ripples beyond those who directly take advantage of the discounts.

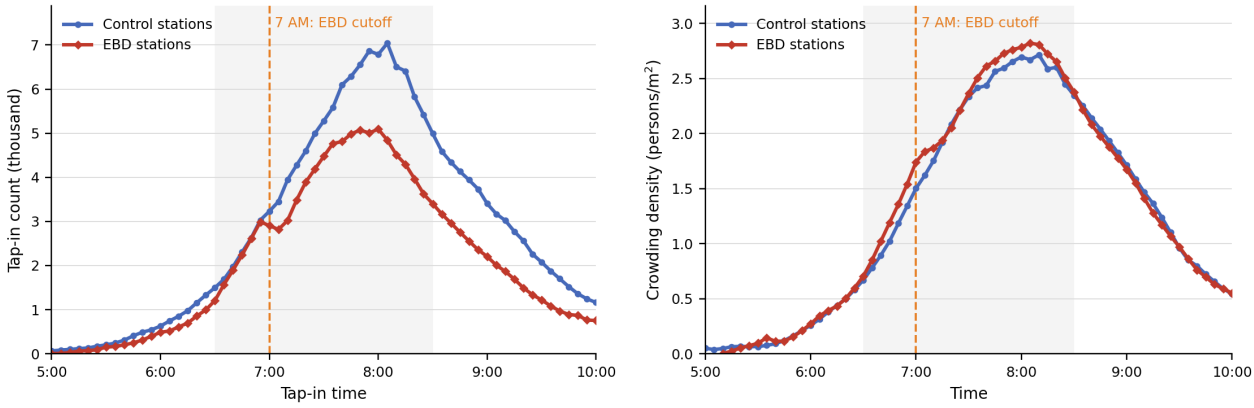
2.2 Data

The primary data source is trip-level records collected by Beijing Subway's automated fare collection (AFC) system (Beijing Municipal Institute of City Planning and Design, 2015). The Beijing Subway uses an electronic smartcard as its payment method. The smartcard is popular because it is easy to use and only card users qualify for fare discounts. By comparing official ridership statistics with our data, we estimate that between 90% and 95% of subway trips were paid for by smartcard around the time of the EBD. In our data, each record captures the tap-in and tap-out stations and timestamps for a single trip, and trips can be linked across days through an anonymized card identifier. The main data used in this paper consist of the universe of smartcard trips on 21 working days in five non-consecutive collection periods in 2016.[4]

We supplement the AFC records with several additional data sources. We geocode all subway stations and collect the track distance and fare for each station pair from Beijing Subway's website. We collect train frequencies by line and station, as well as the types of passenger cars and the number

[4]The collection periods fall in March, June, September, November, and December 2016. June 9–10 and September 15–16 are excluded because of the Dragon Boat Festival and Mid-Autumn Festival holidays, respectively.

Figure 1: Tap-ins and In-Train Crowding during Morning Hours



(a) Tap-in counts

(b) In-train crowding density

Notes: This figure plots, by 5-minute time bin, the number of passengers tapping in (panel a, in thousands) and the in-train crowding density on downtown-bound trains (panel b, persons per square meter) at EBD stations (red diamonds) and control stations (blue circles), averaged over the sample dates. The orange dashed vertical line marks the 7 AM EBD cutoff, and the shaded band marks the 6:30 to 8:30 AM sample period used in the analysis. Control stations show a smooth morning peak in both series. At EBD stations, tap-ins bunch just before 7 AM and dip immediately after, as riders advance their departures to qualify for the discount; the crowding density rises faster than at control stations before the cutoff and flattens just after it, a smaller discontinuity because passengers on the train boarded at upstream stations.

of cars per train, from which we compute the total floor area of each train. For train floor area, we do not distinguish between seating and standing areas. We also use the 2015 Beijing Household Travel Survey (Beijing Transport Institute, 2015), a large representative survey of approximately 100,000 households that records each respondent's home and work locations, daily travel details, and transportation mode choices. We further use grid-level commuting flow data from Baidu Maps to construct OD-pair-specific income distributions. The data and our matching procedure are described in Section 4.1.3. Appendix A provides a detailed description of all data sources used in this paper.

3 Model

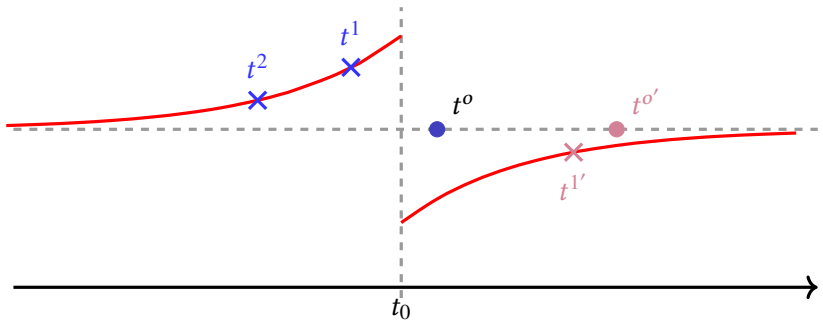
This section presents a model of travel demand in which subway passengers choose their departure time, trading off the fare, expected in-train crowding, and the cost of deviating from an optimal schedule. We first describe the model and then provide an overview of the estimation strategy.

3.1 Model of Subway Demand

Figure 2 illustrates the intuition behind our estimation. For simplicity, suppose the density of travel demand is flat over time, as indicated by the horizontal gray dashed line. Now consider the introduction of the EBD before the cutoff time t_0. Take a passenger who originally planned to depart

right after 7 AM, at t^o. Once the discount is in place, they have an incentive to tap in before the cutoff to claim the lower fare. If the cost of rescheduling increases with the distance from the original departure time, they have no reason to move any earlier than necessary and would depart just before the cutoff, at t^1. However, other passengers may reason the same way, making the train right before the cutoff extremely crowded. Given how much they are willing to pay to avoid crowding, the passenger may instead choose to depart at an earlier time, at t^2. The EBD can also generate ripple effects on those who do not directly benefit from the discount. Consider another passenger who planned to leave at $t^{o'}$. They may now choose to depart earlier, at $t^{1'}$, not to qualify for the EBD but to take advantage of the less crowded subway cars. The red curves trace the likely profile of trip density in the presence of the EBD. The network structure of the subway further complicates passengers' calculations, as those who respond to the EBD generate spillover effects that propagate through the network.

Figure 2: The Departure Time Trade-off under the Early-Bird Discount



Notes: This figure illustrates passengers' departure time choices under the early bird discount.

We model passengers' departure-time choices, taking their origin–destination (OD) pair and route as given. Departure times are discretized into 15-minute bins between 6:30 and 8:30 AM, measured by the tap-in time at the origin station. We define a *market* as an OD pair j on a calendar date d, and each departure-time bin $t \in \mathcal{T}$ is a *product*. The outside option represents traveling between the same origin and destination during morning rush hours by other modes.

Passenger i's utility from departing at time t in market (j, d) is

$$U_{ijdt} = V_{ijdt} + \varepsilon_{ijdt}, \tag{1}$$

where the error term, ε_{ijdt}, is i.i.d. following a Type-I extreme value distribution, and the deterministic component is

$$V_{ijdt} = \alpha_i P_{jdt} + \beta_i \, \mathbb{E}_d[\text{Crowd}_{jdt}] + \rho_i \left| t + \mathbb{E}_d[\text{TT}_{jdt}] - t_{ijd}^{OA} \right| + \kappa_{jd} + \phi_{dt} + \xi_{jdt}. \tag{2}$$

The fare P_{jdt} varies across OD pairs as a step function of distance and varies across time bins within

an OD pair at EBD stations, where passengers receive a 30% discount for tapping in before 7 AM.

On each date d, passengers form expectations about crowding and travel time before choosing when to depart. The operator $\mathbb{E}_d[\cdot]$ denotes these expectations. The term $\mathbb{E}_d[\text{Crowd}_{jdt}]$ is the expected total in-train crowding experienced during the trip, measured in passengers · minutes per square meter; it aggregates crowding density over the duration of the passenger's in-train exposure. We proxy expected crowding with $\widehat{\text{Crowd}}_{j,\text{dow},t}$, where dow denotes the day of the week corresponding to date d. These fitted values come from a regression of realized crowding on OD-by-day-of-week-by-time-bin fixed effects, equivalent to averages across dates within each cell. They capture the systematic conditions that experienced commuters can anticipate.

The rescheduling term $\left|t + \mathbb{E}_d[\text{TT}_{jdt}] - t^{OA}_{ijd}\right|$ captures the absolute deviation between the passenger's expected arrival time—the sum of departure time t and expected travel time—and their optimal arrival time t^{OA}_{ijd}. We proxy expected travel time with $\widehat{\text{TT}}_{j,\text{dow},t}$, using the same averaging procedure as for crowding. We assume the penalty from deviating from the optimal arrival time is symmetric and linear in time.[5]

The remaining terms are OD-by-date (market) fixed effects κ_{jd}, which absorb the common value of completing trip j on date d; date-by-time-bin fixed effects ϕ_{dt}, which capture departure-time patterns common across OD pairs on each date; and a structural demand shock ξ_{jdt} that varies at the OD-by-date-by-time-bin level and is unobserved by the econometrician. The coefficients α_i, β_i, and ρ_i represent the marginal (dis)utility of fare, crowding, and schedule deviation, respectively.

First consider the simple case where preference is homogeneous: $\alpha_i = \alpha$, $\beta_i = \beta$, and $\rho_i = \rho$. Normalizing the outside option utility to zero, the market share of time bin t in market (j, d) takes the standard logit form:

$$S_{jdt} = \frac{\exp(\delta_{jdt})}{1 + \sum_{\tau \in \mathcal{T}} \exp(\delta_{jd\tau})}, \tag{3}$$

where $\delta_{jdt} \equiv \alpha P_{jdt} + \beta \widehat{\text{Crowd}}_{j,\text{dow},t} + \rho \left|t + \widehat{\text{TT}}_{j,\text{dow},t} - t^{OA}_{jdt}\right| + \kappa_{jd} + \phi_{dt} + \xi_{jdt}$ is the mean utility, where t^{OA}_{jdt} is the mean schedule deviation of passengers choosing the same product. Following Berry (1994), the share equation can be inverted to yield the linear estimating equation:

$$\log S_{jdt} - \log S_{jd0} = \alpha P_{jdt} + \beta \widehat{\text{Crowd}}_{j,\text{dow},t} + \rho \left|t + \widehat{\text{TT}}_{j,\text{dow},t} - t^{OA}_{jdt}\right| + \kappa_{jd} + \phi_{dt} + \xi_{jdt}, \tag{4}$$

Because $\log S_{jd0}$ varies only at the market level, it is absorbed by κ_{jd}. Price, the expected crowding experience, and the expected schedule deviation all enter the utility function in Equation (1) in levels, so ratios β/α and ρ/α give the marginal willingness to pay (MWTP), in Chinese yuan (RMB), to avoid one additional passenger per square meter for one minute and the MWTP to reduce schedule deviation by one minute, respectively.

[5]The symmetry and linearity assumptions can be relaxed. We maintain them here for two main reasons. First, the EBD incentivizes riders to reschedule to an earlier departure time. Second, as we show below, the deviation in travel time is small in general, so allowing for nonlinear effects does not make a substantial difference.

Passengers with different income levels may have different price elasticities and derive different levels of disutility from crowding and rescheduling. To capture this heterogeneity, we allow preferences to vary with income:

$$\alpha_i = \alpha_0 + \alpha_1 I_i, \qquad \beta_i = \beta_0 + \beta_1 I_i, \qquad \rho_i = \rho_0 + \rho_1 I_i, \tag{5}$$

where I_i is the passenger income, recentered at the sample mean. Because the smartcard records do not contain passenger income, we simulate income from an OD-specific distribution of potential commuters.[6] With heterogeneous preferences, the utility decomposes into a mean component δ_{jdt}, common to all passengers in the market, and an individual deviation μ_{ijdt} that depends on income:

$$\begin{aligned} V_{ijdt} = & \underbrace{\alpha_0 P_{jdt} + \beta_0 \widehat{\text{Crowd}}_{j,\text{dow},t} + \rho_0 \left| t + \widehat{\text{TT}}_{j,\text{dow},t} - t^{OA}_{jdt} \right| + \kappa_{jd} + \phi_{dt} + \xi_{jdt}}_{\delta_{jdt}} \\ & + \underbrace{\alpha_1 I_i P_{jdt} + \beta_1 I_i \widehat{\text{Crowd}}_{j,\text{dow},t} + \rho_1 I_i \left| t + \widehat{\text{TT}}_{j,\text{dow},t} - t^{OA}_{jdt} \right|}_{\mu_{ijdt}}. \end{aligned} \tag{6}$$

The random-coefficient model no longer allows for a closed-form expression for market shares, and we estimate it using the conventional iterative procedure in the BLP literature (Berry et al., 1995): for a given vector of non-linear parameters $\theta = (\alpha_1, \beta_1, \rho_1)$, predicted shares are computed by simulation over income draws, the mean-utility vector δ is recovered via the contraction mapping, and the resulting structural residuals are interacted with a set of instruments to form the GMM objective.

3.2 Estimation Strategy Overview

Estimating the model requires measuring crowding and optimal arrival times and identifying passengers' responses to fares, crowding, and rescheduling. We first describe how we construct the unobserved variables, then explain the variation and assumptions used to identify the preference parameters. Section 4 presents the detailed measurement procedures and instrument construction.

3.2.1 Measurement

The automated fare collection (AFC) records cover subway trips throughout the network, recording each passenger's origin and destination stations and tap-in and tap-out times. They do not reveal which route a passenger takes, which train they board, or the crowding they experience. We therefore reconstruct passengers' routes and locations during their trips. We develop an iterative

[6] We use Baidu Maps commuting flows and income brackets to construct an income distribution for potential commuters in each OD pair. We fit a beta distribution and draw 150 incomes for each OD-date market. These draws represent potential commuters rather than only observed subway riders; the model allows their choice between subway travel and the outside option to depend on income. Section 4.1.3 describes the construction.

algorithm that jointly estimates the optimal route from the origin to the destination and the time spent walking, waiting, and transferring at each station, with in-train time imputed from the published timetable. Dividing the estimated number of passengers sharing a train by its available floor area yields crowding density. Summing this density across trip segments, weighted by time spent on each segment, gives total in-train crowding exposure.

The optimal arrival time is also unobserved. Passengers choose departure times partly based on their preferred arrival times, and omitting this variable would confound the crowding coefficient: OD-time combinations with clustered optimal arrival times would exhibit both high crowding and high unobserved demand, biasing the estimate of β. We infer each passenger's optimal arrival time from "coworkers"—subway passengers who arrive at the same destination around the same time on the same date but originate from stations not directly affected by the EBD. Their arrival distribution reflects arrival choices without the direct fare incentive to shift schedules. Within each destination-date cell, we reweight this distribution to match the income composition of EBD riders and assign each rider the coworker arrival time at the same percentile of the reweighted distribution. Section 4.1.2 describes the procedure.

Both measurement strategies exploit the *network structure* of the transit system. The train-assignment algorithm traces passengers across the subway network to compute crowding on shared segments. The coworker-based imputation identifies riders who converge at the same destination from different parts of the network. As we discuss next, the same network structure also underpins our instrumental variable strategy for crowding.

3.2.2 Identification

In our setting, identification of the fare coefficient α relies on variation induced by the early-bird discount. The policy reduces fares by 30% for trips originating from 16 designated stations before 7 AM. Outside the discount, fares P_{jdt} depend on the track distance between the origin and destination stations. The EBD stations were not randomly selected; they are among the busiest suburban stations. We include OD-by-date fixed effects κ_{jd} to absorb differences in demand levels across OD pairs on each date, so α is identified from within-market variation over departure-time bins. We also include comparison stations on two other suburban lines that exhibit similar ridership patterns but were not covered by the EBD. The comparison stations help identify the date-by-time-bin fixed effects ϕ_{dt}, allowing us to distinguish the response to the EBD fare reduction from changes in demand over the morning.[7] Identification requires that, conditional on the fixed effects and other trip attributes, unobserved demand ξ_{jdt} does not vary systematically with the EBD-induced fare reduction.

Expected crowding $\widehat{\text{Crowd}}_{j,\text{dow},t}$ may remain positively correlated with ξ_{jdt}, even after controlling

[7]The demand model includes price level, so using trips only from EBD stations, the model can still be estimated with OD-by-date and date-by-time-bin fixed effects. But the model is only identified due to the specific functional form. For example, with only EBD stations, the model is unidentified with log price.

for fares, rescheduling costs, and the fixed effects. This demand shock varies jointly across OD pairs j, dates d, and departure-time bins t. A subway ride in a particular OD pair at a particular time is crowded not because passengers enjoy crowded trains, but because traveling in that OD pair at that time has high unobserved value *despite* crowding. For instance, if many residents near the origin station also work near the destination and conventional work hours require arrival by 9 AM, commuting demand concentrates at specific departure times. To the extent that the fixed effects and measured rescheduling costs do not capture these OD-specific departure-time preferences, they generate both high crowding and high unobserved demand. This positive correlation biases the OLS estimate of β toward zero, or can even make it positive, understating the true disutility of crowding.

We instrument for crowding using passengers whose trips respond to different demand shocks but overlap with the focal trip on shared train segments. These passengers travel from other origins to distant destinations, adding to crowding on the shared segments. We call them *incidental companion passengers* (ICPs). We restrict ICPs to infrequent riders and assume that, conditional on the model's controls, their demand shocks are unrelated to the focal trip's demand shock ξ_{jdt}. Our instrument is expected incidental companion time (ICT), which weights each ICP by the duration of shared travel and uses the same expectation procedure as crowding. Section 4.2 describes its construction.

To identify the rescheduling coefficient ρ, we assume that, conditional on the model's controls, $|t + \widehat{\mathrm{TT}}_{j,\mathrm{dow},t} - t^{OA}_{jdt}|$ is uncorrelated with ξ_{jdt}. One concern is that unobserved demand may increase passenger volume and travel time, changing the rescheduling term. We show below that in-train crowding does not meaningfully slow trains and that controlling for in-station crowding does not materially change the crowding valuation estimates. Another concern is that EBD and control riders may differ in their preferred arrival times t^{OA}_{ijd} even for the same destination and date. These differences can affect both the imputed schedule deviation and demand for particular OD-time combinations, inducing correlation with ξ_{jdt} beyond what the fixed effects absorb. We show in Section 4.1.2 that pre-policy data reveal no systematic differences in arrival patterns between EBD and control riders associated with the 7 AM cutoff.

4 Measurement and Identification

4.1 Measurement

4.1.1 Optimal Route, Train Assignment, and Crowding

To measure in-train crowding, we need to determine each passenger's location at every point during their trip. There are two key challenges to this task. First, the AFC data record only each passenger's tap-in and tap-out stations and times, not which trains they board or how long they spend inside stations. We also do not observe trains' real-time locations or the crowding density on

platforms or inside cars.[8] Second, with 19 lines and more than 340 stations, the Beijing Subway is a complex network, and many OD pairs admit multiple feasible routes. We need to determine which route each passenger takes.

Figure 3 illustrates the two objects our procedure recovers for a single trip. Panel (a) shows the decomposition of the trip's travel time into in-station and in-train components along its route, which places the passenger on a specific track segment at each minute of the trip. Panel (b) shows the crowding density on each segment and time bin the passenger passes through and its sum over the trip, the passenger's crowding experience. The rest of this subsection follows the figure: we first describe how the route and the time components in panel (a) are estimated, and then how they are translated into the crowding measures in panel (b).

Optimal route and travel-time decomposition. We assume that passengers take the route with the shortest travel time and identify optimal routes using observed data. Specifically, we define a route $\mathcal{P}_{jdt}$ as an ordered sequence of stations and their associated times,

$$\mathcal{P}_{jdt} = \{S^O(t^O), S_1(t^1), \ldots, S_n^R(t^n), \ldots, S^D(t^D)\},$$

where S^O and S^D denote the entry and exit stations, S_n the pass-through stations, and S^R a transfer station if applicable. We let $\mathcal{R}_{\mathcal{P}_{jdt}}$ represent the set of transfer stations along route $\mathcal{P}_{jdt}$. t in the parentheses indicates the time when the passenger reaches the corresponding station; only the entry time t^O and the exit time t^D are observed.

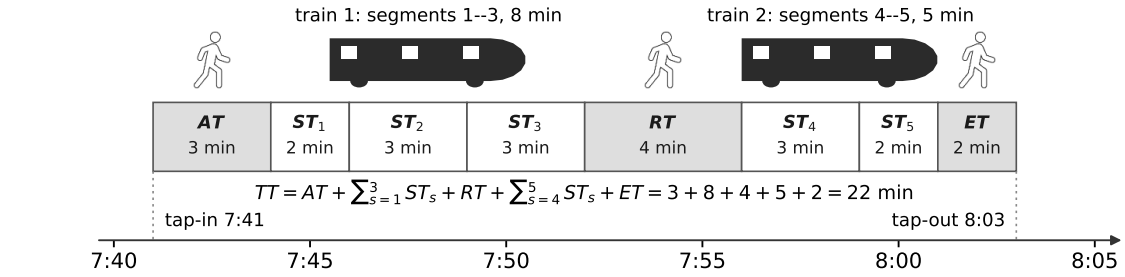
The total travel time along route $\mathcal{P}$ can be decomposed as

$$TT_{jdt}^{\mathcal{P}} = \underbrace{ST_{jdt}^{\mathcal{P}}}_{\text{in-train}} + \underbrace{AT_{t^O}^{O}}_{\text{entry}} + \underbrace{\sum_{R \in \mathcal{R}_{\mathcal{P}}} RT_{t^R}^{R}}_{\text{transfer}} + \underbrace{ET_{t^D}^{D}}_{\text{exit}}. \tag{7}$$

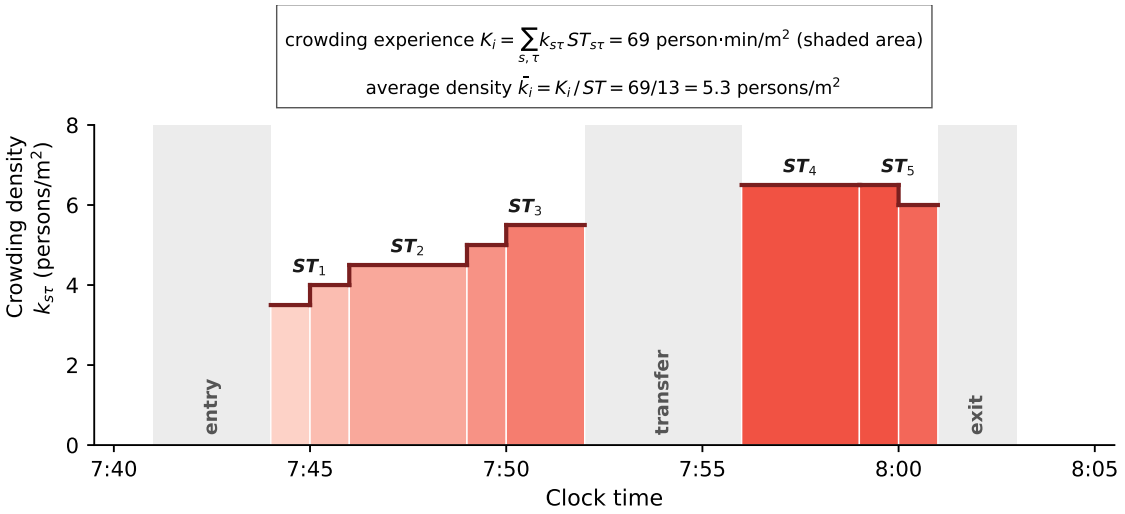
The total travel time is observed from the tap-in and tap-out timestamps. $ST_{jdt}^{\mathcal{P}}$ is the in-train time, the sum of scheduled train running times between adjacent stations along the route. $AT_{t^O}^{O}$ is the entry time between tapping in at the origin station and boarding the first train; $RT_{t^R}^{R}$ is the transfer time at transfer station R when arriving at time t^R; and $ET_{t^D}^{D}$ is the time between leaving the train at the destination station and passing through the exit fare gate. We assume that trains always run on time, an assumption we validate in Section 5.3, so once the route is known, the in-train time follows from the published timetable. The three in-station components are not observed and must be inferred. Panel (a) of Figure 3 shows the decomposition for the illustrative trip: a 22-minute journey from tap-

[8] Most trains in the Beijing Subway are equipped with real-time location devices, which are used to display estimated arrival times at stations. Platforms and passenger cars also have cameras that could potentially be used to derive crowding measures. However, we do not have access to those data.

Figure 3: Trip Decomposition and Crowding Exposure: An Illustration



(a) Travel-time decomposition



(b) Crowding density $k_{s\tau}$ along the trip and crowding experience K_i

Notes: Illustrative crowded morning trip, not sample estimates. The two panels share the clock-time axis, and the ticks delimit the five-minute bins τ. As in Equation (7), TT is the total travel time from tap-in to tap-out, AT the entry time, RT the transfer time, and ET the exit time; ST_s is the scheduled running time on track segment s, and the five ST_s sum to the in-train time ST; $ST_{s\tau}$ is the part of ST_s that falls in bin τ. All are in minutes. $k_{s\tau}$ is the crowding density on segment s in bin τ (persons per square meter), $K_i = \sum_{s,\tau} k_{s\tau} ST_{s\tau}$ the passenger's crowding experience (person-minutes per square meter), and $\bar{k}_i = K_i/ST$ the average density over the in-train minutes. Panel (a) decomposes total travel time, with in-train stages in white and in-station stages in gray: the passenger taps in at 7:41, boards the first train after 3 minutes of entry time, rides segments 1 to 3 (2, 3, and 3 minutes from the timetable), spends 4 minutes transferring at station R, rides segments 4 and 5 on a second train (3 and 2 minutes), and taps out at 8:03 after 2 minutes of exit time. Panel (b) plots the crowding density the passenger is exposed to along the trip. Each shaded piece is one segment-by-bin cell (s, τ), labeled by segment (ST_1 to ST_5), with width equal to the minutes spent in it and height equal to $k_{s\tau}$, computed from all passengers aboard that segment in that bin divided by the train floor area, including passengers whose trips this one never meets. Where a bin boundary falls inside a segment (7:45, 7:50, and 8:00), the segment is split into two cells. The entry, transfer, and exit intervals are shaded gray; in-station crowding is not measured. The shaded area under the profile is K_i, 69 person-minutes per square meter, an average of 5.3 persons per square meter over the 13 in-train minutes.

in at 7:41 to tap-out at 8:03 consists of 3 minutes of entry time, 13 minutes on five track segments taken from the timetable, 4 minutes of transfer at station R between the two trains, and 2 minutes of exit time.

Estimating in-station times and optimal routes. We estimate the in-station components and the optimal route jointly through an iterative procedure applied to the entire subway network. Starting from initial guesses for each in-station component, we compute the total travel time of every candidate route for each OD pair and select the fastest route with Dijkstra's algorithm. Given the routes, we update the in-station components by regressing the observed median total travel time $\hat{TT}_{jdt^O}$ for trips in each OD-tap-in-time bundle on in-train time $\hat{ST}_{jdt^O}$ and a set of station-by-purpose (entry, exit, transfer) fixed effects capturing in-station time cost:

$$\hat{TT}_{jdt^O} = \hat{ST}_{jdt^O} + AT^O_{t^O} + \sum_{R \in \mathcal{R}} RT^R_{t^R} + ET^D_{t^D} + \varepsilon_{jdt^O}. \tag{8}$$

A station can serve as an entry, exit, or transfer station for different trips, and we allow a different time cost in each station-time-bin bundle for each purpose. The regression is estimated by ordinary least squares (OLS), pooling across the entire subway network and weighting by the number of trips in each jdt^O. The number of unique OD pairs is orders of magnitude larger than the number of unique station-purpose pairs, so there are enough degrees of freedom to flexibly identify entry, exit, and transfer times specific to each station. We recalculate the optimal routes and in-train times using the updated estimates and iterate until all in-station time parameters and the implied route choices converge.[9]

Given a trip's optimal route, Equation (8) faces a collinearity problem. The total travel time is observed, the in-train time is determined by the timetable, and the error term is mean-zero, so the in-station time components must sum to a constant equal to the mean total travel time minus in-train time. To break the collinearity, we normalize the mean exit time to two minutes.

Appendix Table C.1 summarizes the estimated in-station times. Entry time, from tapping in at the origin station to boarding the train, averages 3.5 minutes with a standard deviation of 1.84 minutes. This variation may reflect station configurations, train frequencies, and in-station crowding. Exit time averages two minutes by construction, and has a standard deviation of 0.9 minutes. Transfer time, at an average of 4.43 minutes, is the longest in-station component, with some transfer stations requiring more than six minutes. The last column of the table reports the number of observations used to estimate each component. Because all trips have an entry and exit station, these two components are estimated from approximately 33.4 million trips. The average trip involves slightly more than one transfer, so transfer time is estimated from 34.8 million trips.

Not all passengers traveling from O to D with tap-in time t^O arrive at the same time. There is a distribution of total travel time, which we attribute to idiosyncratic factors in the in-station segments. For example, some passengers walk faster than others, some are lucky to catch a departing

[9]While the optimal route may in principle vary over the course of the day due to time-varying in-station congestion, in practice the fastest route for a given OD pair rarely changes by time of day. This is because the second-best routes typically cost substantially more time. We therefore assign the most common optimal route across all departure times to each OD pair.

train while others must wait for the next one, and some may stop at a convenience store or vending machine inside the fare gates. While assuming all passengers take the same optimal route, we treat the sources of travel time variation as agnostic and proportionately scale each in-station component of the median passenger's time cost to fit the observed total time of each individual trip. If the median trip in a bundle has the decomposition in Figure 3(a), a trip in that bundle that takes 24.2 minutes, 10% longer, is assigned 3.3 minutes of entry, 4.4 minutes of transfer, and 2.2 minutes of exit time, while its 13 in-train minutes remain fixed.

The route assignment algorithm rests on three assumptions that merit discussion. First, train speed does not vary with crowding. During the study period, the Beijing Subway was largely new and equipped with advanced computerized control systems. Stopping times at stations are automated, and most stations have platform screen doors to prevent accidents. Unlike some older systems, delays were rare. Section 5.3 shows that the empirical relationship between crowding and travel time is negligible. Second, passengers do not detour to avoid crowded routes. Detours typically cost substantially more time, and Section 5.4 provides evidence that route-switching to avoid crowding is uncommon. Third, the distribution of travel time within an OD-time bundle reflects idiosyncratic shocks and is unrelated to crowding levels.

Crowding density and experience. With each passenger's route and time components in hand, we know their location at every minute of the trip and can turn to panel (b) of Figure 3. We define a *segment* s as the track between two adjacent stations and discretize each trajectory into segment-by-five-minute cells (s, τ). When a passenger traverses several segments within one five-minute bin, their presence is distributed across the segments in proportion to the time spent on each; when a single segment spans two bins, it is partitioned into the corresponding cells. The shaded pieces in panel (b) are the cells the illustrative trip passes through, with widths equal to the minutes spent in each; segments 1, 3, and 5 straddle a bin boundary and are therefore split in two. Summing presence over all passengers yields $N_{s\tau}$, the weighted ridership on segment s during time bin τ.

We focus on in-train crowding.[10] We define the in-train *crowding density* $k_{s\tau}$ as the number of passengers per square meter on segment s in time bin τ, obtained by dividing $N_{s\tau}$ by the total floor area of a six-car train.[11] $k_{s\tau}$ reflects the degree of crowding a passenger experiences at a given time.

[10]Passengers may also derive negative utility from crowded stations. Our route assignment algorithm also allows us to calculate the number of passengers in each station at any given time. We focus on in-train crowding because it is arguably the most important dimension of the disamenity. Including in-station crowding is possible, but because the station is much larger and passenger distribution is uneven, relevant crowding density is difficult to impute. We show in robustness checks that including in-station crowding does not affect the estimates of the WTP to avoid in-train crowding.

[11]Each train in our sample consists of six passenger cars measuring 19 meters long and 2.8 meters wide, providing a total floor area of approximately 320 square meters. We do not subtract floor space occupied by seats and other on-train equipment, nor do we model the probability of finding a seat or differential utilities from sitting and standing. Using smartcard data from the Hong Kong subway similar to ours, Hörcher et al. (2017) presents an algorithm to determine the probability of finding a seat. We do not observe detailed train headways by line and time of day. The ratio of segment run time to headway gives the average number of trains on a segment. Approximate morning-peak headways suggest

The height of each piece in panel (b) is this density; it depends on every passenger aboard that segment in that bin, not only on the focal trip.

For each passenger i, the in-train *crowding experience* aggregates the crowding density over all cells along the route, weighted by the time spent in each:

$$K_i = \sum_{(s,\tau)} k_{s\tau}\, ST_{s\tau},$$

where $ST_{s\tau}$ is the in-train time the passenger spends in cell (s, τ). K_i has units of person-minutes per square meter and captures the total discomfort as the product of the intensity and the duration of crowding exposure. In panel (b) of Figure 3, it is the shaded area under the density profile: 69 person-minutes per square meter, or an average density of 5.3 persons per square meter over the 13 in-train minutes. These trip-level exposures underlie the expected crowding proxy used in the demand model. We aggregate K_i to the product level by computing the mean across all trips within each OD-by-date-by-15-minute-bin (jdt) cell.

4.1.2 Optimal Arrival Time and Rescheduling

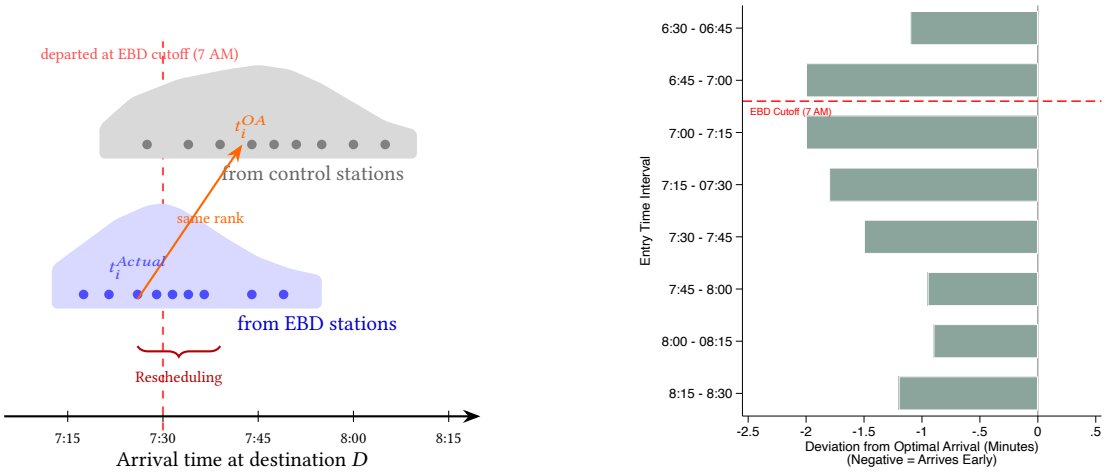
The rescheduling cost in the demand model requires measuring the optimal arrival time t^{OA}_{ijd} for each passenger. This variable is fundamentally unobserved. Observed arrival times may not be optimal because passengers trade off schedule deviation against fare savings and expected crowding. In particular, the EBD incentivizes passengers to depart earlier than they otherwise would, so observed arrival times of trips from EBD stations are also likely to be earlier than optimal. Exploiting the network structure of the subway system, we construct the optimal arrival time using trips that arrive at the same destination station but originated from non-EBD stations.

The key idea is that passengers arriving at the same destination in the morning peak hours as the focal passenger are likely responding to the same need for travel. We refer to these passengers as "coworkers" because they travel to the same destination on the same date and may face similar workplace schedules. Their arrival distribution reflects arrival choices without the direct fare incentive to shift departure times. If control station riders travel to the same destination on the same day as EBD riders, the two groups face similar workplace constraints and destination-side demand conditions. The control riders' arrival distribution therefore serves as a natural counterfactual for what EBD riders would have chosen absent the discount.

Figure 4 panel (a) provides a graphical illustration of this approach. For a given date d and destination station D, we collect all trips arriving at D during the morning rush from control stations. This set of trips forms the control arrival distribution. We then collect the corresponding trips from EBD stations. We further refine the control arrival distribution by reweighting it to match the in-

a ratio of about 0.8–1.2, so using one train's floor area approximates average on-train density when this ratio is close to one.

Figure 4: Optimal Arrival Time Imputation and Validation



(a) Quantile Matching Procedure

(b) Mean Schedule Deviation by Entry Time

Notes: Panel (a) illustrates the quantile matching procedure. The blue distribution shows observed arrivals at destination D from EBD stations, shifted earlier by the fare discount. The gray distribution shows arrivals from control stations traveling to the same destination on the same day. Each EBD rider is matched to the control distribution at the same within-group percentile rank, yielding the imputed optimal arrival time t_i^{OA}. Panel (b) plots the mean signed deviation (actual minus optimal arrival time, in minutes) for EBD station trips by 15-minute entry time bin. Negative values indicate that passengers arrived earlier than the imputed optimal time. The dashed line marks the EBD discount cutoff at 7 AM.

come composition of EBD riders in the same date-destination cell. Because the EBD discount shifts some passengers to earlier departures, the EBD arrival distribution is likely shifted to the left relative to the control distribution, as illustrated in Figure 4 panel (a). We perform quantile matching to map each EBD rider to the control distribution. Specifically, let p_i denote the percentile rank of EBD rider i within the EBD arrival distribution for that date-destination cell. The imputed optimal arrival time is

$$t_i^{OA} = F_{\text{Ctrl}}^{-1}(p_i), \tag{9}$$

where F_{Ctrl}^{-1} is the inverse cumulative distribution function of the income-reweighted control arrival distribution. The matching is performed at high precision using continuous seconds. Cells with fewer than ten EBD trips or fewer than ten control trips are excluded from the procedure.

For trips originating from control stations, we set the optimal arrival time equal to the actual arrival time, $t_{\text{Ctrl}}^{OA} = t_{\text{Ctrl}}^{Actual}$, so that rescheduling cost is zero by construction. This ensures that any nonzero rescheduling cost in the estimation comes from the EBD group, where the fare discount induces measurable departures from schedules not directly unaffected by the EBD.

This approach has several advantages. It relies entirely on contemporaneous data, eliminating concerns about temporal shifts in commuting patterns that would arise from using pre-policy historical records. By matching within the same date-destination cell, it captures date-specific demand

conditions such as weather or local events that affect both EBD and control riders symmetrically. This approach is essentially non-parametric. It assigns coworkers through matching two *distributions*, rather than directly matching individuals. Without knowing the exact workplace for each rider, assigning coworkers to each rider would require more assumptions and likely introduces additional biases.

A key assumption underlying the method is that passengers from EBD and control stations traveling to the same destination face similar optimal arrival incentives. Appendix Figure C.4 provides supporting evidence by comparing the destination distributions of the two groups. Of the 241 unique destination stations in the sample, 188 are served by riders from both EBD and control origins. The remaining 53 single-group destinations account for a negligible share of total ridership (0.55% for EBD-only destinations and 1.21% for control-only destinations). The Pearson correlation between the two destination share vectors is 0.46, and the Spearman rank correlation is 0.68, indicating that the two groups share broadly similar destination patterns despite originating from geographically distinct lines.

Figure 4 panel (b) validates the imputed rescheduling measure by plotting the mean signed deviation (actual minus optimal arrival time) for trips originated from EBD stations across 15-minute entry time bins. Negative signed deviations indicate that the passenger arrived earlier than optimal, which is the expected direction for EBD riders who advance their departures to qualify for the discount. Riders who tap in before the 7 AM cutoff arrive approximately one to two minutes earlier than their imputed optimal time. After 7 AM, when the discount no longer applies, deviations become smaller. The largest deviations are observed for trips that depart around the cutoff. The clear discontinuity at the cutoff confirms that the imputed optimal arrival time captures meaningful variation in rescheduling behavior. The vast majority of the deviations calculated for EBD trips are negative. The rescheduling variable used in the demand model is the absolute deviation between the expected arrival time and the imputed optimal arrival time, $|t + \widehat{\mathrm{TT}}_{j,\mathrm{dow},t} - t^{OA}_{jdt}|$.

Pre-existing differences between EBD and control riders' optimal arrival-time distributions could cause the imputed deviation to reflect scheduling preferences formed before the policy as well as a response to the discount. Applying quantile matching to 2015 data, before the EBD, yields a smooth deviation profile at future-EBD stations with no discontinuity at 7 AM. Comparing the 2016 and 2015 profiles shows an additional advance of 3.3 and 2.9 minutes in the two pre-7 AM bins. This comparison supports interpreting the change near the cutoff as policy-induced rescheduling.

4.1.3 Income Distribution

Passengers may have heterogeneous preferences for trip characteristics. We are particularly interested in preference heterogeneity by income levels, which has implications for the distributional impacts of crowding and crowding-reducing policies. The smartcard data do not have information on user demographics. We match passenger income using novel outside data sources.

We use grid-level commuting flow data provided by Baidu Maps, a leading provider of digital map services in China. Baidu Maps provides location service on smart devices and is embedded in many mobile applications. It is able to track millions of smart mobile devices (mostly smartphones) with precise and high-frequency location data. In particular, it determines a device's usual nighttime location, which it refers to as home, and its usual daytime location, referred to as work. The commuting flow data contains the number of commuters between all pairs of home and work locations in Beijing, where each location is a small grid with an edge of approximately 700 meters, and thus an area of about 0.5 square kilometers. The data are based on the usual location patterns in the three months ending in November 2021 and cover 7.6 million unique devices with home locations distributed across roughly 15,000 grids.[12] Importantly, the data reports the share of commuters in each home-work location pair in five monthly income brackets. User income is inferred by Baidu Maps based on the information it has access to on the device using a machine learning algorithm. The information includes the make and model of the device, applications installed on the device, some information users voluntarily disclosed to Baidu and its partnering applications, and detailed location information. We do not have access to those underlying user characteristics, the exact algorithm Baidu uses, or the training data it uses to validate the imputations. Because there is no official statistics available in China that provide income information at granular geographic levels, the Baidu data is arguably the best such data available. We use the imputed income distribution to estimate the heterogeneous willingness-to-pay to avoid crowding and rescheduling. However, results based on such imputed data should be interpreted with caution.

We assign commuters from the Baidu data to subway station pairs. We define a station's catchment as an area within a three-kilometer radius. Data from the 2015 Beijing Household Travel Survey (BHTS) (Beijing Transport Institute, 2015) show that very few people choose to ride the subway if they live or work more than three kilometers away in straight-line distance from a station. Catchment areas of different stations can overlap. Users in multiple catchment areas are probabilistically assigned to stations with weights equal to the inverse of distance. For example, suppose a worker lives in the catchment areas of both stations S_1 and S_2, the distance from their home to S_1 is 1 km while that to S_2 is 0.5 km, then we assign the worker to S_1 with a probability of 1/3 and to S_2 with a probability of 2/3. Similarly, the worker's workplace can be probabilistically assigned to a station. Applying the same algorithm to all commuters in the Baidu data, we define the user pool of each station pair OD. The number of potential commuters is not equal to the number of trips, as people do not necessarily commute every day and there are non-commuting trips. We use the number of subway trips observed in the smartcard data and the share of subway trips from the BHTS data to convert the user pool into market sizes.

For each OD pair, we estimate a continuous income distribution from the grouped income data

[12] Beijing covers a land area of 16,000 square kilometers and had a population of about 22 million in 2016. The grids do not cover the mountainous regions where few people live and bodies of water.

by fitting a beta distribution via grouped maximum likelihood estimations. The income data is from 2021, while we deflate the incomes to the 2014 price levels, we are still essentially assuming no real income growth and the OD-specific income distribution is time-invariant. For the random-coefficient demand model we estimate below, we draw 150 simulated passenger incomes from the fitted distribution for each OD-date market.

Appendix Figure C.5 plots the distribution of simulated monthly incomes. The weighted mean income is approximately 8,770 RMB per month. The median monthly income of potential subway riders is about 8,000 RMB, and the 90th percentile is around 18,000 RMB. Those numbers are broadly consistent with official statistics. The municipal statistics bureau reported an average monthly salary of 7,706 RMB in 2016. Appendix Figure C.6 maps the mean monthly income by residential and work-place location. High-income residents are spread across suburban areas, while high-income work-places concentrate along the central business corridors. These geographic patterns are consistent with conventional knowledge of spatial residential and job distributions in Beijing.

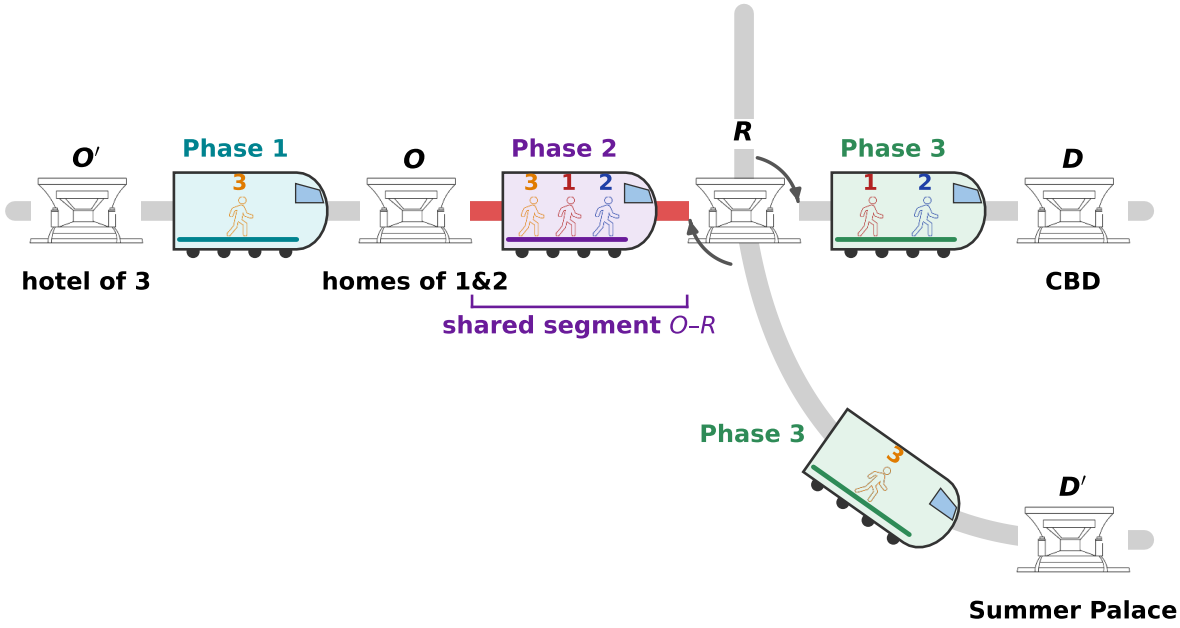
We estimate a multinomial logit mode-choice model on the BHTS with individual income and trip distance as predictors, and apply the fitted shares to the Baidu commuting population. Appendix Figure B.1 reports mode choices as functions of income and travel distance. Subway shares rise sharply with trip distance, especially among those with high income. More than 40% of commuters earning above the sample median (8,000 RMB) choosing the subway for commutes longer than 12 km. Cars are primarily used by the high-income for trips of intermediate distances, while bus is used mostly for short trips. Appendix B provides the full estimation results and the predicted mode shares by income and distance.

4.2 Instrument for Crowding

Crowding is endogenous because the demand shocks of passengers who ride the same train are correlated. Figure 5 illustrates the problem and our solution on a stylized network with two lines crossing at the transfer station R. Passengers 1 and 2 live near O and work in the central business district (CBD) at D, where they must be by 8 AM. The shared work schedule raises their demand for the same departure time, and because they board the same train at O, the train is crowded exactly when their demand is high (Phase 2). The model's fixed effects do not absorb this shock: κ_{jd} absorbs the value of the OD pair on the date and ϕ_{dt} the departure-time pattern common to all OD pairs, but the shock is specific to the OD pair *and* the time bin and therefore remains in ξ_{jdt}. Across OD pairs and departure times, high unobserved demand comes with high crowding, so OLS biases the crowding coefficient β upward, toward zero or even positive, and the implied WTP to avoid crowding, β/α, downward, to the point of turning negative in the OLS estimates of Section 6.1.

Identifying β requires crowding generated by passengers whose travel demand is unrelated to the focal market's. Passenger 3 in Figure 5 is such a passenger: a tourist staying at a hotel near

Figure 5: Endogeneity of Crowding and Incidental Companion Passengers: An Illustration



Notes: This figure illustrates the endogeneity of crowding and the incidental companion passengers (ICPs) used to address it, in three phases of one morning on a stylized network with two lines crossing at the transfer station R; the train icons, colored by phase, are snapshots of the same journey. Phase 1: passenger 3, a tourist staying at a hotel near O', boards at O'. Phase 2: passengers 1 and 2, who travel from their homes near O to the CBD at D to be at work by 8 AM, board at O; their shared work schedule raises demand for this departure time and crowds the train, the source of endogeneity. Passenger 3 rides with them on the segment from O to R (in red). Phase 3: passengers 1 and 2 continue to D, while passenger 3 changes lines at R to reach the Summer Palace at D', which holds a 9 AM event every day. Passenger 3 is an ICP of passengers 1 and 2: a different origin, destination, and purpose, so passenger 3 adds to their crowding without sharing their demand shock. The $m^{i,i'}$ minutes they ride together between O and R are passenger 3's contribution to their incidental companion time (ICT).

O' and heading to the Summer Palace at D', which holds a 9 AM event every day. Passenger 3 boards at O' (Phase 1), rides with passengers 1 and 2 from O to R (Phase 2), and changes lines at R (Phase 3). Tourists like passenger 3 are on this train every morning, not the same tourists but a steady flow set by the event's schedule and by Beijing's visitor traffic, neither of which has anything to do with the commuters' work schedule. Tourists are one instance of a broader pattern. Riders whose travel demand arises from something other than the commuters' workplace and work hours, such as patients with a hospital appointment, students heading to a morning lecture across town, and travelers catching an intercity train at Beijing South Station, play the same role. We call such riders *incidental companion passengers* (ICPs): riders who share part of the train ride with the focal passengers but travel for a different purpose. Nor are ICPs a surprise: passengers 1 and 2 see them every morning and weigh the added crowding when choosing when to depart, and this response is what identifies β.

The relevance condition of the instrument holds by construction: ICPs are counted only over

the segment of the ride they share with the focal passenger, so they are part of the crowding that passenger experiences, and the first stage is strong, with Wald F-statistics above 290 (Table 3). The exclusion restriction requires that, conditional on the model's controls, ICPs' demand is unrelated to the focal market's unobserved demand ξ_{jdt}. The condition fails if ICPs respond to the shock that drives the focal passengers' demand. For example, if passenger 3 were a commuter who works near the CBD and must also be there by 8 AM, passenger 3's trip would follow the same work schedule as the trips of passengers 1 and 2, and passenger 3 would not be an ICP. Neither travel purpose nor the demand shock is observed in the smartcard data, so we select ICPs by three observable criteria: rider frequency, origin, and destination.[13] Infrequent riders are unlikely to share the commuters' regular travel pattern, a distant destination rules out a shared workplace schedule and local demand shocks at D, and an upstream origin separates the companion's trip from local conditions at O. Section 6.2 shows how the estimates vary across these criteria. The ICP instrument follows the same logic as the instrument for airline fares in Hortaçsu et al. (2024), which uses demand from other itineraries that share a flight leg: in both settings, the identifying variation comes from travelers who share the same physical capacity but face unrelated demand shocks.[14]

The instrument should match the way crowding enters the model, as expected exposure accumulated over the ride, which takes two steps. First, passengers 1 and 2 have many ICPs, and one who rides along for longer adds more to their crowding experience, so we weight each ICP by the shared travel time. In the notation of the model, passenger 1 or 2 is a focal passenger i traveling from O to D in market (j, d) at tap-in time t; let $\mathcal{I}_i'$ be the set of ICPs of passenger i and $m^{i,i'}$ the minutes shared with ICP i'. The *incidental companion time* (ICT) of the trip is

$$ICT_{ijdt} = \sum_{i' \in \mathcal{I}_i'} m^{i,i'}, \tag{10}$$

in minutes; passenger 3 contributes the $m^{i,i'}$ minutes between O and R. Second, passengers 1 and 2 choose when to depart on the basis of the crowding they expect rather than the crowding realized on the day, so the instrument is expected ICT, $\widehat{ICT}_{j,\text{dow},t}$, constructed exactly as expected crowding in Section 3.1: we compute ICT for each OD-date-departure-time cell and average across dates with the same day of the week.[15]

[13] In the baseline, ICPs are infrequent riders, with no more than three recorded subway trips during the collection week, who board upstream of O and alight at least eight kilometers from D. Infrequent riders account for 6.7% of trips in the estimation sample (Table 1).

[14] Borrowing variation from other markets served by the same firm or capacity has a long history in demand estimation. Hausman (1996) and Nevo (2001) use prices in other markets to capture cost shocks common to the firm, and Fan (2013) uses demand shifters in the other markets a firm serves. Hortaçsu et al. (2024) describe their network instrument as inspired by both and assume that demand on the connecting leg is independent of demand in the focal market, the counterpart of our condition on ICPs.

[15] In implementation, ICT is divided by train floor area before averaging, so the instrument is measured in person-minutes per square meter; this scaling does not change the identifying variation. That variation is, conditional on the OD-by-date and date-by-time-bin fixed effects, the difference in departure-time profiles across OD pairs: ICPs crowd

The random-coefficient model in Equation (5) adds income slopes on fare, crowding, and rescheduling, and pinning down these additional parameters requires more moment conditions than the single ICT instrument provides. We therefore add four instruments based on the EBD policy. The first interacts an indicator for an EBD origin with an indicator for departure before 7 AM. The remaining three interact the EBD-origin indicator with linear, quadratic, and cubic functions of departure time:

$$\text{Treat}_j \times 1\{t < 7{:}00\}, \quad \text{Treat}_j \times t, \quad \text{Treat}_j \times t^2, \quad \text{Treat}_j \times t^3.$$

Here Treat_j indicates an EBD origin and t denotes departure time, as in the demand model. The first instrument captures the discount cutoff, while the remaining instruments capture differences in the departure-time profile between EBD and control origins.

We control for fare P_{jdt}, so these instruments exploit variation in expected crowding that remains after accounting for fare and the model's other controls. Figure 2 illustrates why such variation arises. The discount draws passengers toward departures just before 7 AM, increasing crowding and inducing some passengers to depart earlier within the same discounted-fare period. It also changes crowding after the cutoff, affecting passengers whose fares remain unchanged. These responses produce a crowding profile that differs from the step change in fares. The first-stage estimates in Appendix Table C.5, column (2), show that the policy interactions predict expected crowding conditional on fare, rescheduling, and OD-by-date and date-by-time-bin fixed effects.

Using these interactions as instruments requires that, conditional on the model's controls, they are unrelated to unobserved demand ξ_{jdt}. We use them alongside expected ICT in the multiple-instrument specification and in the random-coefficient model to help identify the income-dependent preference coefficients.

4.3 Sample and Summary Statistics

Our sample includes trips originating from the 16 EBD stations and 16 control stations and ending at any of the 342 stations in operation in 2016. We restrict to trips with a tap-in time between 6:30 and 8:30 AM. The EBD discount applies to passengers who tap in before 7 AM, so the two-hour window spans across the time cutoff. While the first train departs the terminal station around 5 AM on all four lines, ridership is low in the early hours (Figure 1), and we choose the earliest time of the sample at 6:30 AM. The 32-station restriction applies to the focal trips used in demand estimation. We measure crowding and construct ICPs using trips throughout the subway network, subject to the restrictions in Section 4.2.

The sample covers five non-consecutive collection periods spanning late February to December

some time bins from O more than others, and ICPs on other lines crowd different bins. These profiles are stable across dates, which is what makes crowding predictable (Section 5.2); the stability does not weaken the instrument, because no variation across dates is used.

2016. We exclude non-workdays because the EBD does not apply on those dates. These exclusions include June 9–10 for the Dragon Boat Festival and September 15–16 for the Mid-Autumn Festival, leaving 21 working days in the estimation sample.

Our baseline regressions study all riders departing from the selected stations during the morning peak. As a robustness check, we restrict the demand-estimation sample to frequent riders while continuing to construct the instrument from infrequent riders, as defined in Section 4.2. Results are reported in Appendix C. Crowding is always measured using the full universe of rides.

Each market is an OD pair on a date, with eight products corresponding to 15-minute departure-time bins. We retain markets with positive ridership in all eight bins. The final sample contains approximately 3.8 million trips across 40,191 markets. The EBD subsample includes 17,777 markets covering 1,621 unique OD pairs; the control subsample includes 22,414 markets covering 1,920 unique OD pairs. Regressions are weighted by the number of trips in each market.

Table 1: Summary Statistics

	EBD Stations		Control Stations	
Variable	Mean	SD	Mean	SD
# OD pairs	1,621		1,920	
# markets	17,777		22,414	
# passengers	10.96	21.81	12.41	16.28
# frequent passengers	10.23	20.96	11.57	15.62
Distance (km)	23.85	8.67	21.87	7.87
Total travel time (min)	56.33	19.80	47.81	16.16
In-train travel time (min)	39.67	15.18	37.31	14.34
Price (yuan)	5.21	1.15	5.45	0.84
Rescheduling, signed (min)	−0.28	11.01	−0.19	3.61
Rescheduling, absolute (min)	7.81	7.76	2.40	2.70
Crowding K (person·min/m^2)	132.75	65.32	123.25	52.07
Crowding density k (person/m^2)	3.31	1.30	3.44	1.28
Crowding density from ICP (infrequent)	0.08	0.07	0.08	0.08
Crowding density from ICP (total)	0.85	0.79	0.94	0.98
Crowding experience from ICT (infrequent)	1.48	1.26	1.30	1.17
Crowding experience from ICT (total)	14.97	13.99	14.62	14.51

Notes: Each observation is a product (jdt), defined as an OD pair–date–15-minute time bin. There are 142,216 observations in EBD covered OD pairs and 179,312 in control OD pairs. Means and standard deviations are unweighted across products. Crowding and companion measures describe realized exposure; demand estimation uses their expected-value proxies. ICP denotes incidental companion passengers; crowding density from ICP has a unit of persons/m^2. ICT stands for incidental companion time; crowding experience from ICT has a unit of person·min/m^2. "Infrequent" restricts to passengers with no more than three trips in the current week.

Table 1 reports product-level summary statistics for the key variables in the treatment and control

groups. The two groups are broadly comparable. The average number of subway passengers per product is 11 in the EBD sample and 12.4 in the control sample. Trips from both groups are long, averaging 23.9 km and 56.3 minutes for those from EBD stations and 21.9 km and 47.8 minutes for those from control stations. This is not surprising because both EBD and control stations are at the far ends of suburban lines. The average in-train time is 39.7 minutes for trips from EBD stations and 37.3 minutes for those from control stations. The average fare is 5.21 RMB in the EBD sample and 5.45 RMB in the control sample, reflecting both the distance-based pricing schedule and the EBD discount. The signed rescheduling deviation averages −0.28 minutes in the EBD sample (SD 11.01) and −0.19 minutes in the control sample (SD 3.61). The corresponding absolute deviations average 7.81 and 2.40 minutes (SDs 7.76 and 2.70), respectively.[16] The negative mean rescheduling among the EBD sample is consistent with riders advancing their departures to qualify for the discount.

Passengers in the sample experience substantial crowding. The unweighted product-level average in-train crowding density is 3.31 persons per square meter for trips originating from EBD stations and 3.44 for those from control stations. The total in-train crowding experience, which integrates real-time crowding density over the duration of the trip (Figure 3), averages 132.8 person-minutes per square meter for EBD stations and 123.3 for control stations.[17]

ICPs, here defined as those with an origin O' upstream of O, and a destination D' at least eight kilometers away by straight-line distance, contribute meaningfully to crowding. For trips from EBD stations, total ICPs contribute a crowding density of 0.85 persons per square meter, accounting for roughly 26% of measured crowding. Infrequent riders, our preferred set of ICPs, contribute 0.08 persons per square meter. When weighted by the duration of shared travel, the incidental companion time of infrequent riders averages 1.48 person-minutes per square meter. The control sample has comparable figures.

5 Stylized Facts

This section documents several empirical patterns that motivate our modeling and identification choices. We show that the EBD policy effectively creates bunching in departure times and affects crowding, that crowding is predictable from regular commuting patterns, and that in-train crowding does not cause meaningful delays or rerouting. These facts support treating in-train crowding as a pure disamenity and validate our crowding measures, which assumes that passengers take the fastest route.

[16] The model uses absolute deviations based on expected travel time $\widehat{\mathrm{TT}}_{j,\mathrm{dow},t}$. These can be nonzero for control riders even when their imputed optimal arrival time equals their actual arrival time.

[17] Integrated crowding is computed for each trip before aggregation to products. The average of integrated crowding does not generally equal the product of the separately averaged density and duration.

5.1 The EBD Creates Bunching in Departure Time and Crowding

Figure 1 plots tap-in counts and in-train crowding density at EBD and control stations by 5-min time bin during morning hours; Appendix Figure C.1 breaks the flows at each station group into passengers entering, exiting, transferring, and riding through. At EBD stations, the tap-in flow (panel a) displays a noticeable discontinuity around 7 AM. A cluster of passengers tap in just before the cutoff to qualify for the 30% discount, creating a visible dent in entries immediately afterward. This bunching in departure times in turn disturbs in-train crowding around 7 AM. Because passengers on the train boarded at upstream stations and are themselves responding to the EBD cutoff, the resulting discontinuity in crowding density (panel b) is less sharp than that in tap-ins. The diffusion across upstream stations smooths the effect, but it also means that the EBD generates changes in crowding that ripple beyond the immediate neighborhood of the cutoff time. A passenger scheduled for an 8 AM departure who has no interest in taking the EBD may still adjust their departure time because the crowding on their regular trip is altered by other passengers responding to the EBD.

In contrast, control stations exhibit no such discontinuity in either series. Tap-ins rise smoothly through the morning peak with no bunching around 7 AM, confirming that the EBD is the source of the departure-time distortion observed at treated stations.

5.2 Crowding Is Predictable

A key feature of the demand model is that passengers choose departure times based on *expected* crowding. Table 2 shows that most variation in crowding is captured by OD-by-time-bin fixed effects, with little additional explanatory power from conditioning on day of week. Crowding therefore varies little across observed dates for a given OD pair and departure-time bin. This regularity supports the assumption that experienced commuters can anticipate crowding. As described in Section 3.1, we proxy expected crowding by its average within each OD-by-day-of-week-by-time-bin cell.

These descriptive regressions use different fixed effects from the demand model. The demand model includes OD-by-date and date-by-time-bin fixed effects, which absorb daily demand differences across OD pairs and departure-time patterns common across OD pairs on each date. They leave differences in departure-time crowding profiles across OD pairs. We use expected ICT to instrument for expected crowding conditional on these fixed effects, fare, and rescheduling. The corresponding first-stage estimates are reported in Appendix Table C.5.

5.3 Crowding Does Not Slow Down the Train

Excessive time cost is a potential consequence of crowding and a confounding factor in the WTP to avoid crowding. In many transit settings, such as bus networks, high crowding coincides with road congestion, making it difficult to separate disutility from crowding per se from disutility caused

Table 2: Predictability of Crowding and Ridership

	Crowding Experience (K)			Crowding Density (k)		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Crowding						
R^2	0.8301	0.9876	0.9916	0.6511	0.9812	0.9870
Obs. (million)	5.79	5.74	4.96	5.79	5.74	4.96
	Frequent Riders			Infrequent Riders		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: Ridership						
R^2	0.6804	0.9342	0.9449	0.4317	0.5226	0.6281
Obs. (million)	12.53	12.53	11.93	12.53	12.53	11.93
OD FE	X			X		
OD-Time FE		X			X	
OD-DOW-Time FE			X			X

Notes: Panel A reports R^2 from regressions of realized crowding on fixed effects of increasing granularity. Crowding experience K is measured in person·min/m^2 and crowding density k in persons/m^2. Panel B reports analogous regressions for ridership, separately for frequent riders ($\geq$ 4 trips in the current week) and infrequent riders.

by longer travel times. A distinctive feature of the current empirical setting is that in-train crowding does not meaningfully affect train speed, as trains run on dedicated tracks with fixed schedules, and delay is rare. This makes the Beijing Subway an ideal setting for identifying the pure disamenity cost of crowding. We test this premise directly.

We regress individual trip travel time on average in-train crowding density and in-station passenger counts, controlling for OD-by-date and line-time fixed effects. A one-person-per-square-meter increase in in-train crowding density is associated with only 0.27 additional minutes of travel time, relative to an average of about 39 minutes (Appendix Table C.6, column 1). To put the magnitude in perspective, we convert the delay cost into monetary terms. At an assumed value of time of 0.4 RMB per minute, the implied monetary cost is about 0.11 RMB, an order of magnitude smaller than the willingness to pay for reducing discomfort from crowding. Appendix C.2 reports the calculation and additional evidence. Although we do not directly estimate the disutility from in-station crowding, we show in the robustness checks that in-station crowding does not confound the estimated WTP to avoid crowded trains.

5.4 Crowding Does Not Result in Rerouting

The computation of in-train crowding relies on a train assignment algorithm that assigns each passenger to the fastest route. A natural concern is that some passengers may deviate from this

route in response to crowding. If passengers systematically choose longer but less crowded alternatives, the measured crowding on their assigned route would overstate the crowding they actually experience, and the degree of overstatement would increase with crowding levels.

The distributions of travel time are tightly concentrated around the OD-specific mean and remain similar across crowding quartiles (Appendix Figures C.10 and C.11). A one-person-per-square-meter increase in crowding density is associated with a 0.12-minute increase in absolute deviation from the OD-date-time-bin cell mean, relative to a mean absolute deviation of 2.40 minutes (Appendix Table C.7). These patterns support treating substantial crowding-induced rerouting as uncommon. Appendix C.3 presents the detailed evidence.

6 Estimation Results

6.1 Constant Coefficient Model Estimates

Table 3 reports estimates of the constant coefficient logit model in Equation (4). Crowding and ICT in this section refer to expected values, proxied by the OD-by-day-of-week-by-departure-time-bin averages defined in Sections 3.1 and 4.2; table row labels omit "expected". The first instrument set uses ICT alone; the second adds the four EBD policy interactions described in Section 4.2. Both specifications control for fare, rescheduling, OD-by-date fixed effects, and date-by-time-bin fixed effects.

Column (1) presents uninstrumented OLS estimates with OD-by-date and date-by-time-bin fixed effects. The price coefficient is negative and precisely estimated, indicating that passengers are less likely to choose a time bin with a higher fare. The rescheduling coefficient is also negative and statistically significant at conventional levels. The crowding coefficient, however, is *positive*, and the implied MWTP for crowding reduction is correspondingly negative. Rather than reflecting a genuine preference for crowded trains, the "wrong" sign highlights the endogeneity of crowding: trips taken at crowded times and on crowded routes carry high unobserved value. The positive correlation between crowding and unobserved demand confounds the true disutility, biasing the OLS estimate of β upward.

We use expected ICT to instrument for expected crowding, as described in Section 4.2. Our preferred ICT measure uses infrequent ICPs who board upstream and travel to destinations at least 8 km from the focal destination. Section 6.2 examines sensitivity to these restrictions.

The remaining columns of Table 3 report GMM estimates. Columns (2) and (3) use ICT alone and report the first-stage and second-stage results, respectively. Columns (4) and (5) add the four EBD policy instruments and again report the first and second stages. Column (6) uses the same instruments as columns (4) and (5), replacing the linear crowding term with its square.

In the first stage (columns 2 and 4), ICT is a strong positive predictor of crowding, with coefficients

Table 3: Constant Coefficient Model Estimates

	OLS: Uninstrumented	GMM: ICT only		GMM: ICT + EBD instruments		GMM: Nonlinear
		1st Stage	2nd Stage	1st Stage	2nd Stage	2nd Stage
	(1)	(2)	(3)	(4)	(5)	(6)
Crowding	0.0058		−0.0191		−0.0092	
	(0.0003)		(0.0018)		(0.0011)	
Crowding^2						−0.000028
						(0.000002)
Price	−0.2046	−5.0535	−0.3146	29.2149	−0.1754	−0.1457
	(0.0141)	(0.4153)	(0.0182)	(2.0870)	(0.0087)	(0.0084)
Rescheduling	−0.0297	0.0229	−0.0278	−0.1363	−0.0253	−0.0256
	(0.0009)	(0.0278)	(0.0012)	(0.0221)	(0.0009)	(0.0009)
ICT		11.2860		10.9076		
		(0.4580)		(0.4441)		
Other IVs				✓		
First-stage F-statistic		607.1		299.4		479.3
Overall R^2	0.659		0.523		0.608	0.618
OD-Date FE	X	X	X	X	X	X
Date-Time FE	X	X	X	X	X	X
Mean Rescheduling (mins)	4.796		4.796		4.796	4.796
Mean Crowding (person·min/m^2)	127.4		127.4		127.4	127.4
Implied MWTP						
Crowding	−0.0284		0.0607		0.0526	*see*
	(0.00250)		(0.00540)		(0.00597)	*Fig. C.8*
Rescheduling	0.145		0.0882		0.144	0.1755
	(0.0111)		(0.00678)		(0.00883)	(0.0120)

Notes: This table reports OLS and GMM estimates of Equation (4) for the overall sample of riders. Column (1) reports uninstrumented OLS estimates. In columns (2)–(3), crowding is instrumented by ICT from infrequent riders with $|D - D'| \geq 8$ km. In columns (4)–(6), four EBD policy interactions are added as instruments (Treat $\times 1\{t < 7{:}00\}$, Treat $\times$ trend, Treat $\times$ trend^2, Treat $\times$ trend^3). Column (6) includes a quadratic crowding term. Overall R^2 includes the fixed effects and is computed as one minus the ratio of the structural residual sum of squares to the total centered sum of squares of log shares. The F-statistics assess first-stage instrument relevance; column (6) reports the statistic for the quadratic specification. Standard errors in parentheses, clustered at the OD level. MWTP computed as the ratio of the relevant coefficient to the price coefficient. Appendix Table C.2 reports a parallel set of estimates restricted to the sub-sample of frequent riders. Crowding and ICT denote expected values; the row labels omit "expected".

of 11.3 and 10.9 and first-stage Wald F-statistics of 607 and 299 in the ICT-only and ICT-plus-EBD specifications, respectively. The F-statistics comfortably exceed conventional thresholds for weak instruments. The price coefficient in the ICT-plus-EBD first stage (column 4) is much larger in magnitude than in the ICT-only first stage (column 2) because the additional instruments, the EBD policy interactions with time trends, are highly collinear with price. The EBD discount is itself a function of treatment status and time of day, so the policy interactions absorb most of the price variation, leaving the first-stage price coefficient identified off a smaller and more idiosyncratic residual.

Turning to the second stage, the crowding coefficient is negative in all GMM specifications, in sharp contrast to the positive OLS estimate in column (1). With ICT alone (column 3), the crowding coefficient is -0.0191 and the price coefficient is -0.315, yielding an implied MWTP for crowding

reduction of 0.061 RMB per person·min/m^2. The implied MWTP to reduce rescheduling by one minute is 0.0882 RMB. The ICT-plus-EBD specification (column 5) produces a similar implied MWTP of 0.053 RMB for crowding reduction and 0.144 RMB for rescheduling reduction. The ICT-plus-EBD specification is our preferred specification.

To gauge the economic magnitudes, consider a representative trip lasting approximately 38 minutes in train, using the unweighted product-level averages in Table 1. The willingness to pay to reduce crowding density by one person per square meter over the course of the trip is $0.053 \times 38 \approx 2.0$ RMB, or roughly 38% of the average fare of 5.3 RMB. Assuming linear marginal disutility, the total crowding disutility for a trip at the mean total crowding of about 127 person·min/m^2 amounts to $0.053 \times 127 \approx 6.7$ RMB, exceeding the fare itself. These magnitudes confirm that in-train crowding is a first-order determinant of the value of a subway trip. For rescheduling, deviating from the optimal arrival time by 15 minutes costs $0.144 \times 15 \approx 2.2$ RMB. Relative to the median hourly wage of about 43 RMB (7,500 RMB per month over 22 eight-hour workdays), the rescheduling cost of a 15-minute deviation amounts to about 5% of hourly earnings. The relatively small rescheduling cost may be due to the fact that most rescheduling in our setting is departing and arriving earlier than optimal. Departing in the neighborhood of 7 AM, virtually all trips conclude before 9 AM, the start of conventional work hours. Rescheduling to a later arrival time after the start of conventional work hours may incur a larger cost. Nevertheless, those findings suggest that passengers face a meaningful tradeoff between departing at a less crowded time and arriving closer to their preferred schedule.

Column (6) allows for a non-linear effect of crowding by including the quadratic term of the crowding experience. In this specification, we omit the linear crowding term and include the quadratic term only. This essentially assumes that the marginal disutility is zero when crowding density is zero, which is reasonable. The estimated coefficient on the quadratic term is negative and precisely estimated, indicating that crowding disutility is convex and its marginal cost increases with exposure. The MWTP for crowding reduction varies with the level of crowding, as shown in Figure C.8 in Appendix C. At the 25th percentile of expected cumulative crowding exposure, the MWTP is 0.032 RMB per person·min/m^2, rising to 0.049 at the median and 0.086 at the 95th percentile. The increasing pattern implies that the welfare cost of crowding is concentrated on the most congested trips.

Our estimates of the crowding cost are broadly consistent with the transportation literature, although the comparison requires care, since most prior work expresses crowding as a multiplier on the value of in-vehicle time rather than as a direct disutility per unit of density. Stated preference studies find that this multiplier roughly doubles once passengers stand in dense conditions (Wardman and Whelan, 2011) and that the cost is convex in density (Haywood and Koning, 2015), mirroring the increasing marginal disutility in our quadratic specification, and revealed preference studies using smartcard data reach similar conclusions (Hörcher et al., 2017; Yap et al., 2020). Using the assumed value of uncrowded travel time of 0.4 RMB per minute from Appendix C.2, the preferred coefficient

implies a crowding multiplier of about 1.45 at the mean density of 3.4 persons per square meter and 1.66 at the ninetieth percentile of about 5 persons per square meter.

6.2 Robustness

Alternative ICP Definitions. We examine whether the crowding MWTP estimates are sensitive to the definition of ICPs. The potential endogeneity arises from demand shocks correlated at the origin, the destination, or through common travel purpose. To examine these sources, we vary the construction of the instrument by progressively tightening restrictions on which co-riders qualify as ICPs.

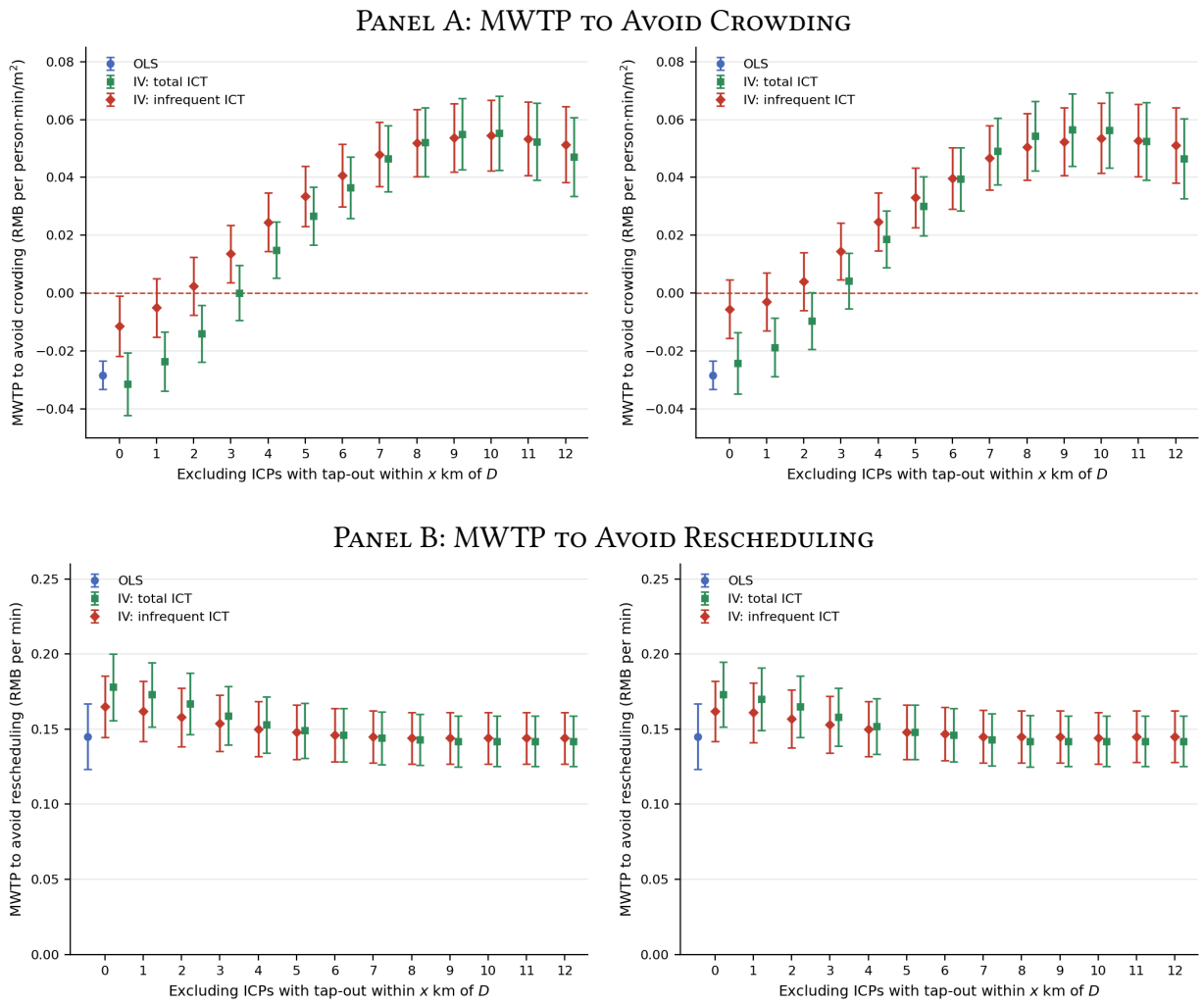
Panel A of Figure 6 compares crowding MWTP estimates across restrictions on the passengers used to construct ICT. Both graphs use ICPs who board upstream of the focal origin O. The horizontal axis increases the minimum straight-line distance between the focal destination D and the ICP destination D'; an 8-km threshold excludes ICPs whose destinations are less than 8 km from D. Graph (b) additionally excludes ICP origins within 2 km of O. The green estimates use all qualifying ICPs, while the red estimates restrict ICPs to infrequent riders. The demand estimation sample remains the overall sample in both cases. All GMM estimates use ICT plus the four EBD policy instruments. The blue dots report the uninstrumented OLS benchmark.

When we only require $D \neq D'$ but distance $|D-D'|$ is not restricted (the leftmost point), the GMM estimate with ICT as IV yields a negative MWTP, with a magnitude similar to the OLS estimate. Using only infrequent ICPs pushes the estimate upward, but the MWTP remains negative and statistically significant. This suggests that ICPs who travel to destinations close to D are likely responding to correlated demand shocks. Restricting to infrequent ICPs improves, but does not fully eliminate the endogeneity concern.

As we gradually remove ICPs whose destinations are close to D, the GMM estimate rises steadily. By restricting $|D - D'|$ to exceed 3 km, the MWTP for crowding reduction becomes positive. Restricting $|D - D'|$ to be greater than approximately 7 to 8 km, the estimates converge. With $|D - D'|$ sufficiently large, estimates using either all ICPs or infrequent ICPs as instruments also converge. This is intuitive. Even when ICPs are traveling for the same purpose (say arriving at work on time), the fact that they are going to a faraway destination means that they are responding to an unrelated demand shock. Thus further restricting to infrequent ICPs does not make a further difference. When $|D - D'|$ is greater than 8 km, the MWTP stabilizes at roughly 0.05 RMB per person·min/m^2. Further tightening the distance restriction does not meaningfully change the estimates.

Graph (b) repeats the exercise while further requiring ICPs to originate from a station O' more than 2 km away from the focal passenger's origin O. The results are nearly identical to those in graph (a), suggesting that the source of endogeneity is not from correlated demand shocks at the origin. Taken together, the two graphs show that the endogeneity of crowding stems primarily from

Figure 6: MWTP across ICP Restrictions



Notes: Each point reports an estimated MWTP from a separate regression on the overall demand sample: Panel A for reducing crowding by one person per square meter for one minute (RMB), Panel B for reducing schedule deviation by one minute (RMB). In each panel, the left graph (a) uses ICPs from all upstream origin stations of O, and the right graph (b) additionally excludes ICPs boarding within 2 km of O. The horizontal axis varies the minimum straight-line distance $|D - D'|$ between focal and ICP destinations. Blue circles report uninstrumented OLS estimates; green squares use expected ICT from all qualifying ICPs; red diamonds use expected ICT from infrequent ICPs (≤ 3 trips in the current week). Both GMM series include the four EBD policy instruments. Vertical bars show 95% confidence intervals. Standard errors are clustered at the OD level.

common shocks at the *destination*. Intuitively, the most important shock that drives crowded subway cars during morning rush hours is the need to arrive at work at a required time, which concentrates travel demand among passengers heading to nearby destinations during the same narrow window.

Panel B repeats the exercise for the MWTP to avoid rescheduling. The estimates stay within 0.14 to 0.18 RMB per minute across all restrictions and close to the OLS estimate, so the correlated demand shock loads on crowding rather than on rescheduling.

These convergence patterns support the preferred 8-km destination restriction used in Table 3.

This restriction creates an exclusion zone of roughly 200 square kilometers around the focal destination. One may worry that it leaves too few qualifying ICPs to generate a strong first stage. This concern is mitigated by the scale of the Beijing Subway network. Figure C.7 in Appendix C shows that the EBD stations are at the far suburban ends of the network, while the vast majority of the system extends well beyond the 8-km radius from the city center (panel a). Because destinations are spread throughout this extensive network, the 8-km exclusion zone around each focal destination D removes only a small share of potential ICPs. Indeed, the median infrequent ICP alights at a station roughly 11 km from D (panel b), well beyond the exclusion threshold. As a result, the first-stage relationship remains strong, with Wald F-statistics exceeding 290 in all specifications (Table 3).

ICP as Instrument. Our preferred instrument uses incidental companion *time* (ICT), which weights each ICP by the duration of shared travel. A natural alternative is to use the expected count of ICPs, discarding the time weights. Table C.3 in Appendix C reports both ICP-only and ICP-plus-EBD specifications using ICP. In the first stage, ICP is a strong predictor of crowding, with a coefficient of approximately 179 and Wald F-statistics of 488.7 (ICP-only) and 302.9 (ICP-plus-EBD). The MWTP for crowding reduction is 0.057 RMB per person·min/m^2 in the ICP-only specification and 0.048 under ICP-plus-EBD. These estimates are close to their ICT counterparts (0.061 and 0.053, respectively), as expected because ICP and ICT exploit the same source of exogenous variation. The stability of the crowding WTP across the two instrument constructions is reassuring.

Frequent riders. The baseline demand model is estimated on trips from all users, while the instrument is constructed from the expected incidental companion time of infrequent riders alone. As a robustness check, we estimate the demand model on frequent riders while continuing to construct the instrument from infrequent riders. This checks whether the estimates depend on including infrequent riders in the demand sample.

Appendix Table C.2 reports the parallel set of estimates. The implied MWTP for crowding reduction is 0.061 RMB per person·min/m^2 with ICT alone and 0.052 with ICT plus the EBD instruments, essentially identical to the overall-sample values of 0.061 and 0.053 in Table 3. We retain the overall demand sample for the remainder of the paper.

Controlling for In-Station Crowding. A potential concern is that in-train crowding may be correlated with in-station crowding, which could independently affect route choice. In-station crowding is qualitatively different from in-train crowding: passengers experience crowded stations in areas where the per-capita space is substantially larger than inside a train car. The main channel through which station congestion could plausibly affect riders is via delays in gates, corridors, and platforms; Section 5.3 shows directly that any such delay is small in magnitude in our setting (a 35 percent increase in in-station passenger counts is associated with about 0.80 additional minutes of travel time;

Appendix C.2). We use a unified definition of in-station crowding: InStn_{jdt} is the total number of passengers that the focal passenger choosing product jdt encounters in stations during entry, transfer, and exit, weighted by the duration of the overlap, and has a unit of thousand persons-minutes.

Table C.4 reports two specifications that add in-station crowding to the estimating equation for the frequent-rider sample. Column (1) adds InStn_{jdt} as an uninstrumented control under the ICT-only specification, while Column (2) treats InStn_{jdt} as a second endogenous regressor, jointly instrumented with in-train crowding by the multiple IVs we have. The EBD policy instruments shift entry timing, which moves both in-train and in-station crowding through the same exogenous mechanism, providing the variation needed to identify both endogenous regressors. The Wald F-statistic is 470.6. The estimated MWTP to reduce in-train crowding remains positive after controlling for in-station crowding. The MWTP to reduce crowding by one person·min/m^2 is 0.05 RMB in Column (1) and 0.0482 RMB in Column (2), compared with 0.061 and 0.052 in the corresponding frequent-rider baseline specifications.

In Column (1), the in-station crowding coefficient is positive (+0.040). A positive coefficient on a disamenity implies the presence of endogeneity: high-demand markets attract more riders, leading to crowded stations and biasing the InStn_{jdt} estimate upward. In Column (2), the in-station coefficient turns negative, and the implied MWTP for in-station crowding reduction is about 0.048 RMB per thousand person·min reduction. Evaluated at the robustness-sample mean (14.3 thousand person·min), the implied disutility from average in-station crowding is roughly $0.048 \times 14.3 \approx 0.69$ RMB, an order of magnitude smaller than the average in-train disutility computed in Section 6.1.

6.3 Heterogeneous Willingness to Pay

The standard logit model restricts all passengers to share the same preference parameters. We now relax this assumption and allow the marginal utility of price, crowding, and rescheduling to vary with income, as specified in the random coefficient model of Equation (5). Each preference coefficient is specified as a linear function of simulated passenger income, recentered at the sample mean of 8.77 thousand RMB. The model is estimated by GMM with the same ICT-plus-EBD instrument set used in the preferred specification, following the BLP contraction mapping procedure (Berry et al., 1995).

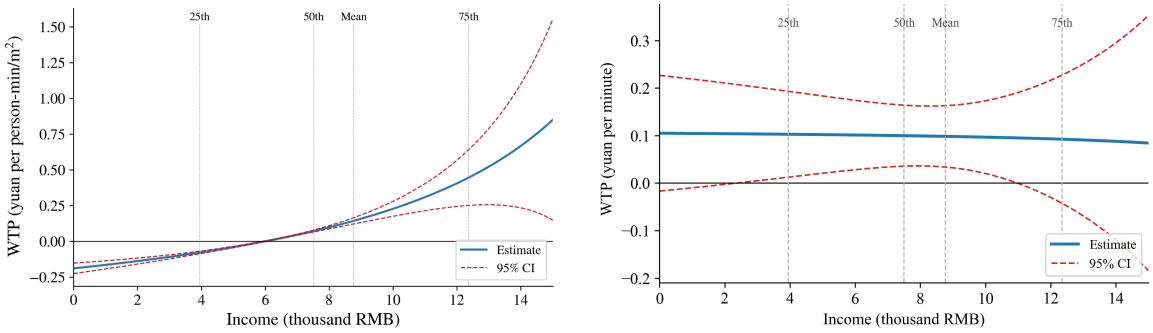
Table 4 reports the estimation results. The price intercept $\hat{\alpha}_0$ = -0.179 indicates that the average passenger derives disutility from fare, while the positive income interaction $\hat{\alpha}_1$ = 0.013 implies that higher-income riders are less price sensitive. The crowding intercept $\hat{\beta}_0$ = -0.026 indicates that crowding reduces utility, and the negative income interaction $\hat{\beta}_1$ = -0.009 reveals that higher-income passengers experience substantially stronger crowding disutility. The rescheduling coefficient $\hat{\rho}_0$ = -0.0176 is negative, but the income interaction $\hat{\rho}_1$ = 0.0015 is not statistically different from zero, suggesting that the cost of deviating from the optimal arrival time is relatively uniform across income levels.

Table 4: Random Coefficient Model Estimates

	(1)
Price (α_0)	−0.1790
	(0.0109)
Price × Income (α_1)	+0.0130
	(0.0047)
Crowding (β_0)	−0.0256
	(0.0006)
Crowding × Income (β_1)	−0.0093
	(0.0002)
Rescheduling (ρ_0)	−0.0176
	(0.0058)
Rescheduling × Income (ρ_1)	+0.0015
	(0.0019)
GMM Objective	0.3030
OD-Date FE	X
Date-Time FE	X

Notes: This table reports the random coefficient model estimates of Equation (5). Income is measured in thousands of RMB per month and recentered at the sample mean (8.77 thousand RMB). The estimates use the overall sample with OD-by-date and date-by-time-bin fixed effects. Heteroskedasticity-robust standard errors in parentheses. The model is estimated using the ICT-plus-EBD instrument set (ICT from infrequent riders with $|D - D'| \geq 8$ km, plus four EBD policy interactions). Crowding and ICT denote expected values; the row labels omit "expected".

Figure 7: Willingness to Pay by Income



(a) WTP for Crowding Reduction

(b) WTP for Rescheduling Avoidance

Notes: The solid line shows the point estimate of MWTP as a function of monthly income. Dashed lines indicate the 95% confidence interval (delta method). Vertical dashed lines mark selected percentiles of the income distribution. Panel (a) reports the MWTP for reducing in-train crowding by one person·min/m^2 (in RMB). Panel (b) reports the MWTP for reducing schedule deviation by one minute (in RMB). The plots are truncated at 15 thousand RMB (between the 75th and 90th percentiles), beyond which the price coefficient $\alpha(I)$ approaches zero and the WTP ratio becomes imprecisely estimated.

Figure 7 plots the estimated MWTP for crowding reduction and rescheduling avoidance as a function of monthly income, with 95 percent confidence intervals computed by the delta method. The MWTP for crowding reduction is steeply increasing in income (panel a). At the median income of 7.5 thousand RMB, the MWTP is approximately 0.07 RMB per person·min/m^2, close to the estimate in the homogeneous setting of 0.053 from Table 3. At the 90th percentile of income (17.5 thousand RMB), the MWTP rises to roughly 1.62 RMB per person·min/m^2, more than twenty times the median value.[18]

In contrast, the MWTP for rescheduling avoidance is relatively flat across income levels (panel b), hovering around 0.10 RMB per minute throughout most of the income distribution. The stability of the rescheduling WTP is consistent with the insignificant income interaction $\hat{\rho}_1$ and suggests that the cost of arriving at an inconvenient time is not strongly differentiated by income.

The steep income gradient in crowding WTP has important implications for policy. It suggests that higher-income passengers would be willing to pay a substantial premium for less crowded service, while lower-income passengers are relatively indifferent to crowding. This pattern provides a potential rationale for service differentiation, where higher-income riders self-select into a more expensive but less crowded option. We explore this possibility in the welfare analysis of Section 7.

7 Policy Counterfactuals

Subway crowding is a congestion externality. Each additional rider imposes discomfort on fellow passengers that is not incorporated in the fare. In this section, we compare three policies that address this externality: a Pigouvian crowding tax, capacity rationing through queuing, and a two-class train configuration with differentiated fares.

We evaluate these policies using a simplified version of the estimated demand model. We replace the subway network with a single origin–destination pair and collapse the morning peak into one representative departure period. Passengers choose between the subway and an outside option, with ridership and crowding determined in equilibrium. The model retains differences in fare sensitivity and crowding valuations across income groups but abstracts from departure-time adjustments and interactions across routes. These simplifications avoid solving simultaneously for passenger choices and crowding throughout the network and across departure periods.

The simplified model preserves the mechanisms underlying the policy comparison. A crowding tax makes passengers pay for the discomfort they impose on others. Queuing limits ridership through waiting and schedule-disruption costs. Differentiated fares and car densities allow passengers to sort

[18]The graph also shows that the MWTP to reduce crowding turns negative for passengers with monthly income under 6,000 RMB, or about 40th percentile of the income distribution. The substantial share of users with negative MWTP and the steep gradient with regard to income are mostly due to the underlying income data, which is imputed by Baidu Maps with limited and indirect information. In welfare calculation and counterfactual analysis below, we define high- and low-income passengers from income percentiles to avoid negative or very high values of MWTP.

according to their crowding valuations. The quantitative effects nevertheless depend on the margins available to passengers. In the full setting, passengers may respond by traveling at another time or taking another route; here, they can leave the subway or, under the two-class policy, switch classes. The exercise therefore illustrates the aggregate and distributional effects of these policy mechanisms, rather than determining optimal fares across the network and over time.

7.1 Setup

Consider a stylized linear city where all residents live in location O and work in location D. O and D are connected by a subway line and a surface road. There are two income groups $g \in \{\ell, h\}$ that equally divide the entire population, $M_\ell = M_h = M/2$, where M denotes the total market size. Each resident commutes to work. They either ride the subway or take the outside option, the utility from which is standardized to one.

Deterministic utility from choosing the subway is

$$V_g = A + \alpha_g P + \beta_g \underbrace{\frac{\text{TravelTime}}{\text{CarArea} \times J}}_{L} N, \tag{11}$$

where A is a common mean utility, P is the fare, N is the total subway ridership on the segment, which is an endogenous outcome of the equlibrium. Let $L = \text{TravelTime}/(\text{CarArea} \times J)$, where J is the number of cars in a train, thus the denominator indicates the total floor space in a train. NL measures the total crowding exposure a passenger experiences during the ride (crowding density times minutes). In this simple static setting, commuters travel at the same time, so departure from optimal arrival time does not enter the utility. We relax this assumption when we introduce queuing in the quantity control policy experiment. Passengers derive disutility from price and crowding, $\alpha_g < 0$ and $\beta_g < 0$. Idiosyncratic taste shocks are i.i.d. type-I extreme value, so the logit share for group g is $s_g(V_g) = e^{V_g}/(1 + e^{V_g})$ and group ridership is $N_g = M_g\, s_g(V_g)$, with $N = \sum_g N_g$.

We split riders at the 41st and 59th percentiles of the simulated income distribution from the Baidu commuting data.[19] The low-income group has a monthly income of $y_\ell = 6.19$ thousand RMB, while the high-income group has $y_h = 9.11$ thousand RMB. The BLP estimates in Table 4 imply $(\alpha_\ell, \beta_\ell) = (-0.213, -0.002)$ for the low-income group and $(\alpha_h, \beta_h) = (-0.174, -0.029)$ for the high-income group.

Baseline ridership follows directly from the data. The ridership-weighted average crowding density is 4.03 persons per square meter, car area is 53.2 m^2, and a train has $J = 6$ cars, yielding an equilibrium ridership of $N_{\text{eq}} = 4.03 \times 53.2 \times 6 = 1{,}286$. Given a ridership-weighted subway mode

[19]The split p41/p59 is symmetric around the median (41 + 59 = 100). The 41st percentile is chosen because it sits just above the point at which the crowding coefficient changes sign, so that low-income riders have a weakly non-negative crowding coefficient while high-income riders strictly dislike crowding. Because low-income riders defined in our case derive little disutility from crowding, and because the MWTP increases sharply

share of 0.43 (constructed from the BHTS imputation on our working sample), the market size is M = 1,286 / 0.43 ≈ 2,983. Ridership-weighted average in-train travel time of 31.83 minutes gives L = 31.83 / (53.2 × 6) = 0.1. The ridership-weighted baseline fare is P_{base} = 5.08 RMB, which we assume is the marginal operating cost. The mean utility A is calibrated so that the private equilibrium at P_{base} reproduces N_{eq}, which yields A = 1.77. Table 5 reports parameter values used in the welfare and counterfactual exercises.

Table 5: Parameter Values Used in the Counterfactual Simulations

Parameter	Value	Description
Structural parameters (from Table 4)		
α_0	−0.179	Price coefficient, intercept
α_1	+0.013	Price coefficient, income slope
β_0	−0.026	Crowding coefficient, intercept
β_1	−0.009	Crowding coefficient, income slope
ρ_0	−0.017	Rescheduling coefficient, intercept
ρ_1	+0.002	Rescheduling coefficient, income slope
Group-level parameters, evaluated at demeaned incomes		
α_ℓ, α_h	−0.213, −0.174	Price sensitivity by group
β_ℓ, β_h	−0.002, −0.029	Crowding disutility by group
Market and calibration		
$y_\ell, y_h, \bar{y}$	6.19, 9.11, 8.77 (k RMB)	Group incomes (p41/p59) and sample mean
N_{eq}	1,286	Baseline equilibrium ridership
M	2,983	Total market size
L	0.100	Congestion exposure (min/m^2/person)
P_{base}	5.08 RMB	Baseline fare
A	1.77	Calibrated mean utility
Two-class configuration		
J_B, J_S	3, 3	Business, Standard car count
λ	0.5	Nesting parameter

Notes: Structural parameters are from the random-coefficient logit estimates in Table 4. Group-level coefficients are evaluated at demeaned incomes $y_g - \bar{y}$ where $\bar{y}$ = 8.77 thousand RMB is the sample mean. Baseline ridership, market size, and congestion exposure are computed from the sample. The mean utility A is calibrated to reproduce N_{eq} at P_{base}.

For the two-class counterfactual we partition the six-car train into 3 Business and 3 Standard cars. The symmetric split is a natural benchmark because $L_B = L_S = 2L$ and the two classes are identical apart from the fare. Riding in either class has the same mean utility A. Class-specific utilities are $V_{gB} = A + \alpha_g P_B + \beta_g L_B N_B$ and $V_{gS} = A + \alpha_g P_S + \beta_g L_S N_S$. Any crowding difference between the two classes is therefore driven entirely by self-selection in response to the fare gap, not by a mechanical

capacity asymmetry or differences in other features.

Through the well-known independence of irrelevant alternatives (IIA) property, a standard multinomial logit would mechanically inflate the subway share once the two-class configuration is introduced. This is because even if prices are identical, draws of idiosyncratic preferences over the two subway classes draw passengers into each subway class. We therefore use a nested logit, where the two classes of subway cars form the lower nest, while subway and other modes form the upper nest. The inclusive value for group g is

$$\mathrm{IV}_g \;=\; \lambda \,\ln\!\Big(e^{V_{gB}/\lambda} + e^{V_{gS}/\lambda}\Big), \tag{12}$$

and the corresponding choice probabilities follow the standard nested-logit form with nesting parameter $\lambda \in (0, 1]$. When $\lambda = 1$, the model collapses to multinomial logit. When $\lambda < 1$, Business and Standard classes are more correlated with each other than either is with the outside option, which attenuates the spurious variety bonus. We use $\lambda = 0.5$ as our preferred value, at the lower end of the range typical for sub-modes within a transit nest in the discrete-choice transport literature, and examine sensitivity to $\lambda \in [0.3, 0.8]$ in the appendix.

7.2 Private Demand, Social Demand, and Externality

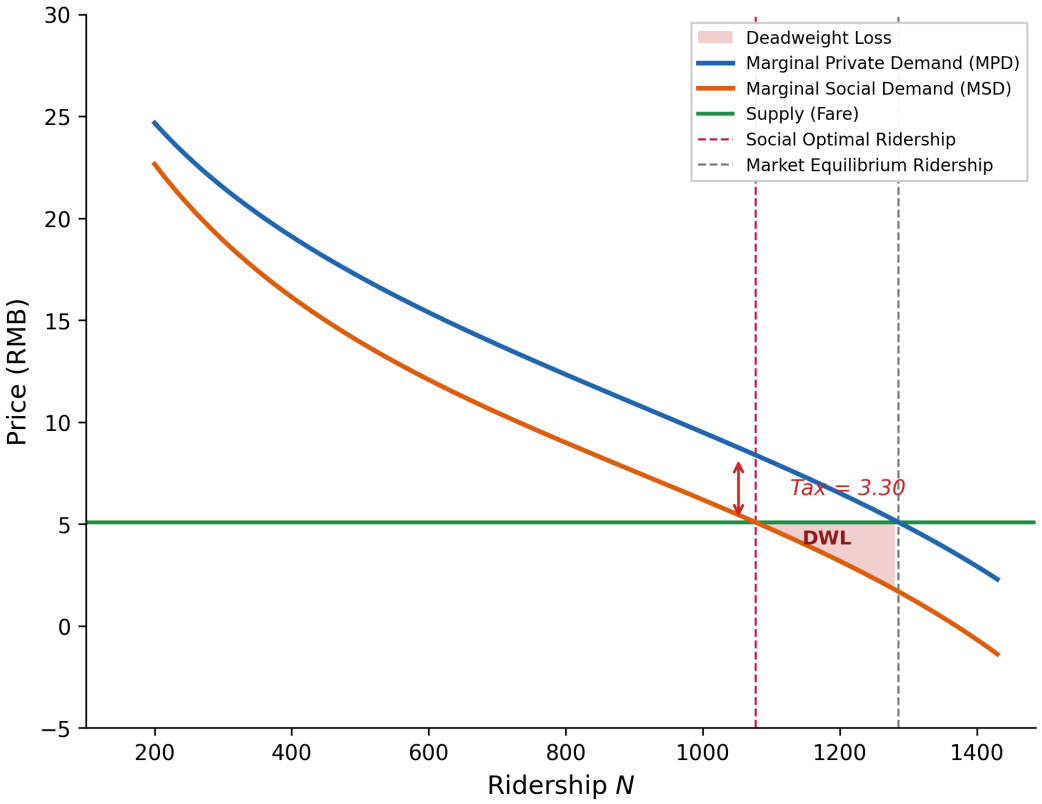
The welfare analysis builds on the distinction between marginal private demand and marginal social demand, illustrated in Figure 8. In the market equilibrium, the price equals each rider's marginal willingness to pay net of the crowding disutility the rider experiences personally. Each additional passenger, however, also imposes crowding on every other rider, a cost the private choice does not internalize. The social planner accounts for this externality, and the vertical distance between private demand and social demand captures the marginal externality.

To construct the inverse private demand, fix any candidate aggregate ridership N. The price that rationalizes N as a private equilibrium must satisfy condition $N = \sum_g M_g\, s_g(A + \alpha_g P + \beta_g L N)$, which implicitly defines the mapping $PD(N)$. Letting $N_g(P, N) = M_g\, s_g(A + \alpha_g P + \beta_g L N)$ denote group-specific ridership, the difference between private and social demand, thus the negative externality, can be written as

$$w(N) \;\equiv\; PD(N) - MSD(N) \;=\; L \sum_{g \in \{\ell, h\}} \frac{\beta_g}{\alpha_g}\, N_g\big(PD(N), N\big). \tag{13}$$

The intuition behind Equation (13) is straightforward. Each additional rider raises crowding linearly in L, and the monetized disutility per unit of crowding for group g is β_g/α_g. Summing these monetized effects across groups, weighted by the number of riders exposed, gives the aggregate marginal external cost.

Figure 8: Crowding Externality and the Optimal Crowding Tax



Notes: MPD denotes marginal private demand, MSD marginal social demand, and the vertical distance between them is the externality wedge. At the social optimum N^*, the Pigouvian crowding tax equals the wedge $w(N^*)$, and the deadweight loss from unpriced crowding is the shaded triangle between MPD and MSD over $[N^*, N_{\text{eq}}]$.

7.3 Counterfactual Policies

We evaluate three policy instruments against the baseline: a Pigouvian crowding tax, capacity-rationing queuing, and a two-class pricing scheme.

We assume P_{base} is the operator's marginal cost, and the operator cannot operate at a loss. The constrained social optimum solves

$$PD(N^*) - w(N^*) \;=\; P_{\text{base}}, \tag{14}$$

and the corresponding Pigouvian crowding tax equals the externality at the socially optimal level of ridership, $T^* = w(N^*)$. Riders face $P_{\text{cons}} = P_{\text{base}} + T^*$. At P_{cons}, individual choices under the competitive equilibrium achieves the social optimal levels. Tax revenue is $T^* N^*$.

Capacity-rationing queuing is a non-price instrument in which each train has a fixed boarding capacity equal to N^* and excess demand produces a queue of length W in minutes. The wait enters utility through two channels: as a time cost priced at the group-specific value of time VOT_g, which

we take as half of the wage rate, and as a schedule-disruption cost captured by the rescheduling coefficient ρ_g estimated in Section 6.3. The generalized price is $P_{\text{base}} + \text{VOT}_g \times W$, and utility includes an additional term $\rho_g \times W$. The equilibrium wait W^* clears the queue, $\sum_g M_g\, s_g(V_g(W^*)) = N^*$, and we solve by bisection on W. Queuing generates no transferable revenue, and the wait cost is an additional deadweight loss; users decide to join the queue with rational expectations about the equilibrium waiting time.

The two-class counterfactual maintains revenue neutrality $P_B N_B + P_S N_S = P_{\text{base}}\, N_{\text{eq}}$. For any given P_S, we find $P_B > P_S$ and solve the nested-logit fixed point in (N_B, N_S), and retain the pair that maximizes the variety-adjusted consumer surplus defined in Section 7.4. The numerical implementation is described in Appendix D.

7.4 Welfare Calculation

We now describe how consumer surplus and total welfare are computed for the baseline and three counterfactuals introduced above. Because the idiosyncratic taste shocks are type-I extreme value, consumer surplus in each group admits the standard log-sum form. In the single-class case, which includes the baseline, the Pigouvian tax, and the queuing counterfactual,

$$CS_g(N, P) \;=\; -\frac{M_g}{\alpha_g}\,\ln\!\Big(1 + e^{A+\alpha_g P+\beta_g LN}\Big), \qquad CS = \sum_g CS_g. \tag{15}$$

The factor $1/\alpha_g$ converts utility into monetary units, and the crowding externality is already embedded in V_g. In the two-class nested logit, the log-sum argument is replaced by the inclusive value, giving $CS_g = -(M_g/\alpha_g)\ln(1 + e^{\text{IV}_g})$.

The nested logit formula mechanically exceeds single-class consumer surplus even when $P_B = P_S$, because the inclusive value exceeds the deterministic utility according to Jensen's inequality. This artificial variety bonus is a known feature of discrete-choice models, which tend to overstate the welfare gains from expanding the choice set when the additional alternatives do not correspond to genuine product differentiation. Because relabeling physically identical cars as Business or Standard does not constitute real variety, we remove this artifact. Let VB denote the consumer-surplus gain when we impose the revenue-neutral equal price, $P_B = P_S = P_{\text{eq}}$. The variety-adjusted consumer surplus is

$$CS_{\text{VA}} \;=\; CS_{\text{total}} - \text{VB}, \tag{16}$$

which by construction equals baseline consumer surplus in the absence of price differentiation. We report CS_{VA} as the primary welfare measure for two-class counterfactuals. Total welfare is reported as $W = CS + \text{TaxRev}$, with CS_{VA} substituting for CS in the two-class case. Under revenue neutrality, fare revenue is held fixed across counterfactuals.

7.5 External Road Congestion

Changes in subway ridership also affect surface-road congestion as displaced riders shift to other modes. Let $\text{MECC}_{\text{veh-km}}$ denote the marginal external cost of road congestion per vehicle-kilometer, $\bar{d}$ the mean trip distance in kilometers, and κ the average car occupancy, so that the passenger-level congestion cost in cars is $c_{\text{car}} = \text{MECC}_{\text{veh-km}}/\kappa$. Diversion shares to other modes differ across groups. We assume that low-income riders shift to $p_{\text{bus},\ell} = 0.5$, $p_{\text{car},\ell} = 0.25$, and other modes with probability 0.25, while high-income riders shift to $p_{\text{bus},h} = 0.15$, $p_{\text{car},h} = 0.75$, and other modes with probability 0.10, reflecting the much higher rate of car ownership among higher-income households. Following Gu et al. (2024), each bus per kilometer generates 0.4 car-equivalents of congestion, while riders shifting to other modes are assumed not to affect road traffic. The change in road-congestion cost relative to baseline is

$$\Delta\text{MECC} \;=\; c_{\text{car}}\,\bar{d}\,\sum_{g}\left(p_{\text{car},g} + 0.4\,p_{\text{bus},g}\right)\left(N_g^{\text{base}} - N_g^{\text{cf}}\right). \tag{17}$$

We report welfare both without and with MECC, as W and $W - \Delta\text{MECC}$.

7.6 Counterfactual Results

Table 6 summarizes the comparison across policies. Panel A reports prices and ridership, and Panel B reports consumer surplus, revenue, and welfare.

The Pigouvian tax, at 3.3 RMB per trip, lowers ridership to the socially optimal level of $N^* = 1{,}077$, a 16 percent reduction from baseline. Consumer surplus falls from 10,885 to 7,671 RMB, but the tax generates 3,555 RMB in transferable revenue, so total welfare rises to 11,226 before accounting for road congestion. Once road-congestion costs from displaced riders are deducted, welfare is 10,737, slightly below baseline. The incidence is sharply regressive. Low-income consumer surplus falls by 3,237 RMB, while high-income consumer surplus actually rises by 23 RMB, because higher-income riders gain nearly as much from the reduction in crowding as they lose from the higher fare. Even when tax revenue is redistributed by ridership share, the net welfare change is −707 RMB for low-income riders and +559 for high-income riders.

Capacity-rationing queuing also targets N^*, with an equilibrium wait of $W^* = 7.75$ minutes, which is about 2.58 headways at the train frequency of 3 minutes. Consumer surplus is 7,782 RMB. Because the instrument generates no revenue, total welfare stays close to consumer surplus and far below the Pigouvian benchmark. The queuing deadweight loss, computed as $\sum_g N_g\, W^*(\text{VOT}_g + |\rho_g/\alpha_g|)$, is approximately 3,434 RMB, almost exactly the revenue that the Pigouvian tax would collect. This equivalence is the classical result from the queuing: rationing by time wastes as deadweight loss what rationing by price collects as transferable revenue.

Two-class pricing produces the highest welfare among all four policies. Revenue neutrality at λ

Table 6: Welfare Comparison across Counterfactual Scenarios

Panel A: Prices and Ridership

	P	tax	$W^\star$	N	N_ℓ	N_h	k
1. Single-class configuration							
1.1 Baseline (status quo)	5.08	–	–	1,286	1,132	154	4.03
1.2 Optimal crowding tax	8.38	3.30	–	1,077	919	158	3.37
1.3 Quantity control (queuing)	5.08	–	7.75	1,077	938	139	3.37
2. Two-class (3+3, $\lambda = 0.5$)							
2.1 Welfare maximization	–	–	–	1,524	1,235	289	4.77
Business	10.30	–	–	357	69	288	2.24
Standard	2.44	–	–	1,167	1,166	1	7.31

Panel B: Consumer Surplus, Revenue, and Welfare (1,000 RMB)

	Consumer Surplus			Tax	Δ Road Cong.	Welfare		Δ Welfare (w/ MECC)	
	CS	ΔCS_ℓ	ΔCS_h	Rev.	Exter.	w/o MECC	w/ MECC	ΔW_ℓ	ΔW_h
Baseline	10,885	0	0	0	–	10,885	10,885	0	0
Crowding tax	7,671	-3,237	23	3,555	489	11,226	10,737	-707	559
Queuing	7,782	-3,008	-94	0	524	7,782	7,258	-3,469	-158
Two-class	11,737	27	825	0	-820	11,737	12,556	272	1,399

Notes: Panel A reports the consumer price P (equal to $P_{\text{base}} + T^*$ under the crowding tax), tax rate, queue wait W^* (in minutes), ridership totals by group, and crowding density k (persons/m^2). Panel B reports consumer surplus and changes by group, tax revenue, the change in road congestion cost, and welfare with and without the MECC adjustment. The last two columns report group-level welfare changes inclusive of MECC: $\Delta W_g = \Delta CS_g + \text{TaxRev}_g - \text{MECC}_g$, so $\Delta W_\ell + \Delta W_h$ equals the change in welfare with MECC relative to baseline. The queuing scenario targets the social optimum $N^* = 1{,}077$. Two-class pricing uses the variety-adjusted consumer surplus $CS_{\text{VA}} = CS_{\text{total}} - \text{VB}$. The revenue-neutrality target is $P_{\text{base}} \times N_{\text{eq}} = 6{,}533$ RMB. All counterfactuals use $\lambda = 0.5$ and the 3+3 car allocation.

= 0.5 implies optimal fares of $P_B^* = 10.30$ and $P_S^* = 2.44$ RMB. Total ridership rises to 1,524, an 18.5 percent increase over baseline, as the discounted Standard fare attracts new riders from the outside option. Variety-adjusted consumer surplus reaches 11,737 RMB, a 7.82 percent gain. The ridership expansion also reduces road congestion, adding 820 RMB in MECC savings. Variety-adjusted welfare including MECC is 12,556 RMB, 15.4 percent above baseline. In terms of distributional impacts, both groups gain on net once road-congestion benefits are redistributed by group: $\Delta W_\ell = 272$ and $\Delta W_h = 1{,}399$. Two-class pricing is the only instrument under which no group is worse off.

Three lessons emerge from the comparisons. First, transferable revenue is a first-order source of welfare for any policy that reduces ridership. The Pigouvian tax dominates queuing purely because its revenue can be redeployed, but queuing is slightly for progressive. Third, service differentiation combined with revenue-neutral pricing exploits the steep income gradient in crowding WTP documented in Section 6.3. High-income riders self-select into less crowded Business cars at a premium that cross-subsidizes lower Standard fares. Both groups gain on consumer surplus once road-congestion spillovers are included, so two-class pricing achieves a Pareto improvement.

8 Conclusions

This paper provides a revealed-preference estimate of subway passengers' willingness to pay to avoid crowding. Combining smartcard data from the Beijing Subway with an early-bird discount that creates plausibly exogenous variation in fares across departure times, we estimate that the marginal WTP to reduce in-train crowding by one passenger per square meter is about 0.053 RMB per minute. Aggregated over the average trip, this implies a crowding externality that exceeds the fare itself, suggesting that the unpriced congestion cost is a first-order distortion in subway pricing. The willingness to pay rises sharply with income while price sensitivity falls, so the average estimate masks substantial heterogeneity in how passengers value less crowded service. Passengers also dislike deviating from their preferred arrival time, with a baseline MWTP of 0.144 RMB per minute of schedule deviation. In the random coefficient model, this MWTP is about 0.10 RMB at the median income. Unlike crowding, this rescheduling cost is essentially flat across the income distribution, indicating that high- and low-income riders place similar value on arriving on time even as they differ markedly in how much they will pay to avoid a crowded car.

Building on these estimates, we evaluate three policies that target the social optimum. A Pigouvian crowding tax raises welfare relative to the unpriced equilibrium, but the gain is driven entirely by transferable revenue, and the incidence is sharply regressive. Capacity-rationing queuing produces consumer surplus close to the Pigouvian level but underperforms the tax because it generates no revenue, with the queue's wait cost almost exactly equal to the foregone fiscal transfer. A two-class configuration combined with revenue-neutral pricing dominates the alternatives. Higher-income passengers self-select into a more expensive but less crowded Business car, the discounted Standard fare draws new riders from the outside option, and total welfare rises by roughly 15 percent above baseline once road-congestion spillovers are included. Both income groups are strictly better off, so two-class pricing is the only instrument under which no group bears a welfare loss.

Methodologically, the paper contributes to the literature on transit demand and congestion pricing along three dimensions. We use granular smartcard data to construct trip-level measures of in-train crowding from imputed routes, develop a novel instrumental variable based on overlapping trips between unrelated origin-destination pairs that addresses the endogeneity of crowding, and combine demand estimation with a calibrated welfare model that quantifies both within-system surplus and external road-congestion effects. From a policy standpoint, our results suggest that operators of crowded transit systems can achieve large efficiency gains and a Pareto improvement without raising average fares, provided that fare differentiation is accompanied by genuine differentiation in the riding experience.

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Crowding
Online Appendix (Not for Publication)

Yizhen Gu Qu Tang Kai Wu Ben Zou

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A Data Sources

This appendix summarizes the data sources used in the paper.

Beijing Subway smartcard records. The primary data source is the trip-level records from the Beijing Subway's automated fare collection (AFC) system (Beijing Municipal Institute of City Planning and Design, 2015). Each record captures the tap-in and tap-out stations and timestamps for a single trip, and trips can be linked across days through an anonymized card identifier. We estimate that between 90 and 95 percent of subway trips were paid by smartcard around the time of the EBD. The main sample consists of the universe of smartcard trips on 21 workdays in five non-consecutive collection periods in 2016, covering trips originating from the 16 EBD stations and the 16 control stations during the 6:30 to 8:30 AM window.

Subway network and operating information. We geocode all subway stations and collect the track distance and fare for each station pair from the Beijing Subway's official website. We also collect train frequencies by line and station, the types of passenger cars, and the number of cars per train. These inputs are used to compute the total floor area available on each train segment at each moment and to determine the scheduled in-train travel time between adjacent stations. We treat seating and standing areas identically.

2015 Beijing Household Travel Survey. The Beijing Household Travel Survey (Beijing Transport Institute, 2015) is a large representative survey of approximately 100,000 households conducted by the Beijing Transport Institute. It records each respondent's home and work locations, daily travel details, and transportation mode choices. We use the BHTS to fit the multinomial logit mode-choice model used in defining market sizes and the outside option.

Baidu Maps commuting flow data. We obtain grid-level commuting flow data from Baidu Maps, a leading digital map provider in China. The data track usual nighttime (home) and daytime (work) locations of roughly 7.6 million devices across about 15,000 residential grids with an edge of approximately 700 meters. For each home-work grid pair, the data report the share of commuters in each of five monthly income brackets. Income is imputed by Baidu's internal machine-learning algorithm from device characteristics, installed applications, and location patterns. We use the Baidu data to construct OD-specific income distributions of potential subway riders.

Auxiliary sources. We collect supplementary information on the 16 EBD stations and their parallel control stations from the Beijing Subway's public announcements. Official annual ridership statistics used to benchmark smartcard coverage are from the Beijing Municipal Commission of Transport.

B Market Size and Mode Choice Estimation

This appendix describes the multinomial logit mode-choice model fitted to the 2015 BHTS and the construction of OD-pair market sizes used in the demand estimation.

The Baidu commuting flow data capture trips in all modes of transportation. To define market sizes and the outside option in the demand model, we need to predict which commuters use the subway. We fit a multinomial logit model of mode choice using the 2015 BHTS, which includes a large representative sample of households with a rich set of demographic characteristics and mode choices for their trips.[1]

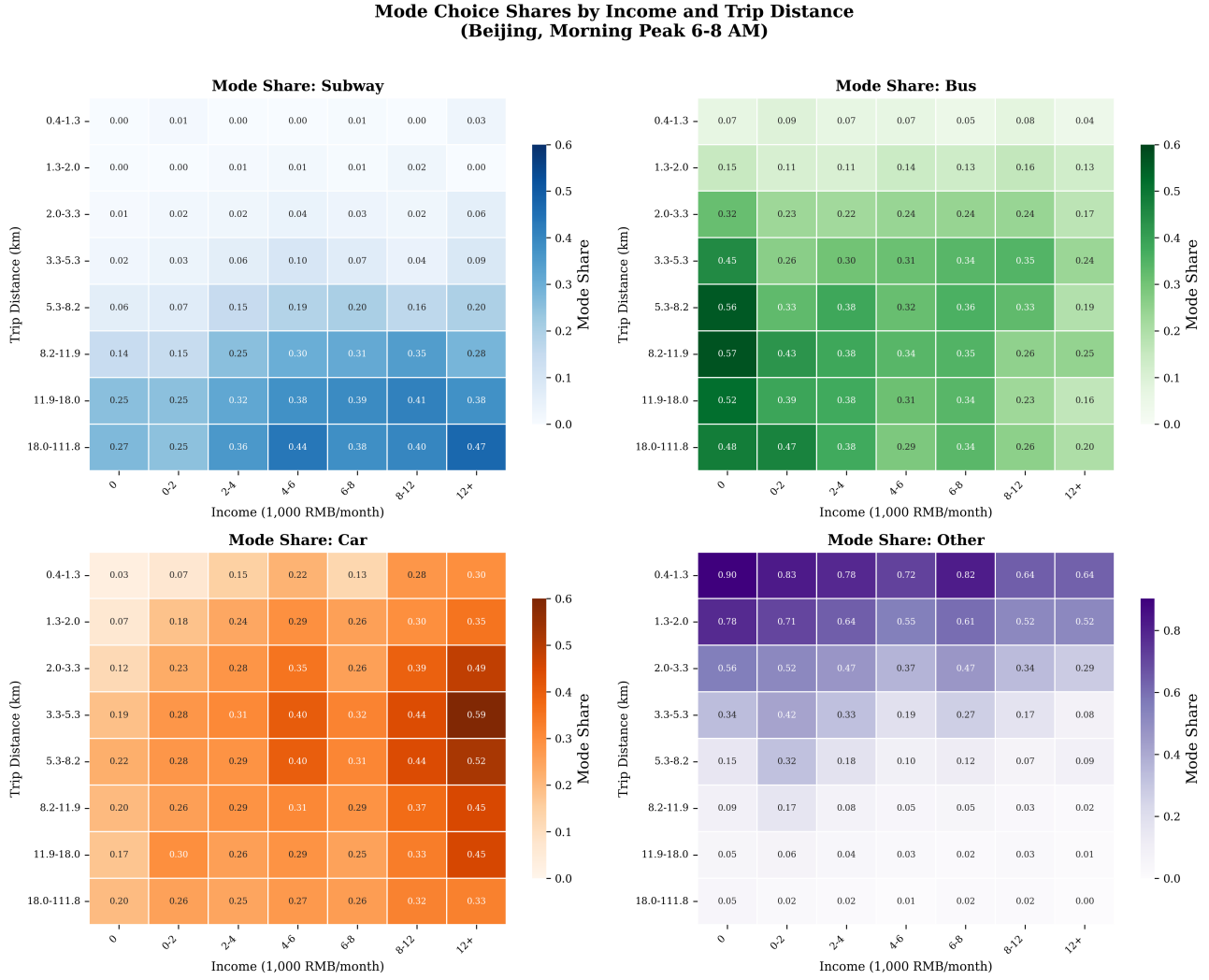
We fit a multinomial logit where each observation is a commuting trip. The dependent variable is the main transportation mode, defined as the mode used for the longest leg. Modes include the subway, bus, and car, with other modes (bicycles, electric scooters, walking) as the other category. Explanatory variables include individual income, its square, and trip distance. Table B.1 reports the estimation results. Higher income is positively associated with all motorized modes relative to walking, biking, or scooting, with the largest effects for subway and car. Trip distance is also strongly positively associated with motorized modes.

Figure B.1 shows mode choice shares from the BHTS by income and trip distance during the morning peak (6 to 8 AM). The subway share rises sharply with trip distance, especially for middle- and high-income commuters. For commuters with an income above 8,000 RMB (the sample median), more than 40% choose subway for commutes longer than 12 km. In contrast, very few choose to ride the subway for trips under 5 km. Bus is primarily used by low-income individuals for middle-to-long distance trips. Car use is concentrated among higher-income commuters and increases with both income and distance, exceeding 30% for high-income commuters on longer trips. Subway nevertheless edges out for very long trips even at the highest income levels. Other modes, primarily walking and cycling, dominate only for short trips under 2 km and decline rapidly as distance grows, regardless of income.

To define market sizes, we use the fitted mode choice model to extrapolate to each OD pair. We predict user mode choices from the Baidu data using predicted mode shares based on income and travel distance. Let $\hat{s}_j^{\text{sub}}$ denote the subway mode share for OD pair j derived from the BHTS. The total market size is $M_{jd} = Q_{jd}^{\text{sub}} / \hat{s}_j^{\text{sub}}$, where Q_{jd}^{sub} is the observed subway ridership from the smartcard data. The outside good encompasses all non-subway modes and is given by $Q_{jd}^{\text{out}} = M_{jd} - Q_{jd}^{\text{sub}}$.

[1] The BHTS reports only household income (in five bins), not individual income. We predict individual income by regressing household income (converted to log RMB) on flexible functions of each working household member's gender, age, education level, industry and occupation, their interactions, and a household fixed effect. The estimated coefficients are used to predict individual income subject to the constraint that all working members' predicted incomes sum to the observed household total.

Figure B.1: Mode Choice Shares by Income and Trip Distance



Notes: Mode shares across income and trip distance bins from the BHTS 2015 survey (morning peak, 6 to 8 AM). Each cell shows the fraction of travelers in that income-distance bin choosing the given mode. Income is monthly household income per capita (1,000 RMB). Distance bins are based on quantiles of reported trip distance.

Table B.1: Estimation of Travel Mode Choice

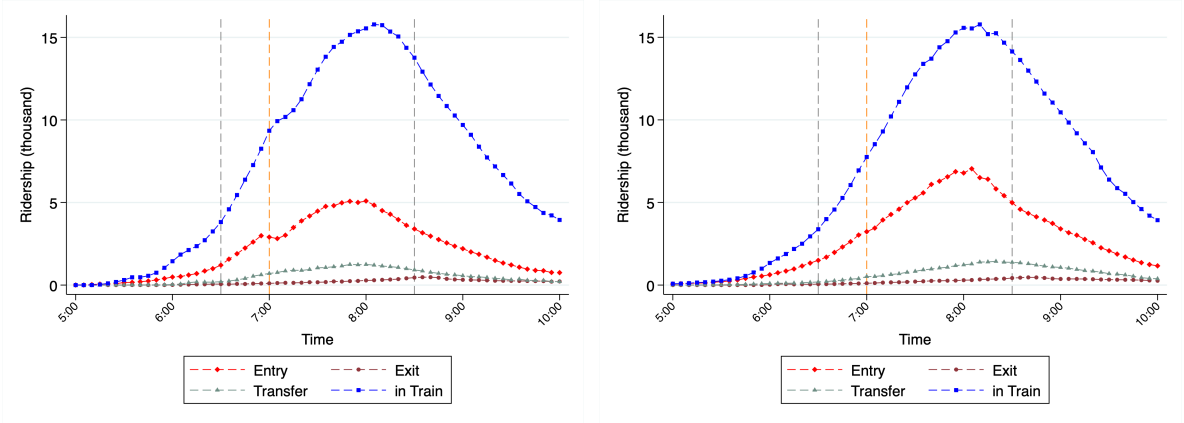
	(1) Subway	(2) Bus	(3) Car
Income (1,000 RMB)	0.271	0.071	0.202
	(0.007)	(0.004)	(0.004)
Income^2	−0.006	−0.001	−0.003
	(0.000)	(0.000)	(0.000)
Trip Distance (km)	0.473	0.437	0.412
	(0.004)	(0.004)	(0.004)
Observations		80,259	
Pseudo R^2		0.258	

Notes: Multinomial logit estimates. The baseline category is "other" modes (bicycles, electric scooters, walking). Standard errors in parentheses.

C Additional Empirical Results

C.1 Passenger Flows and In-Train Crowding Density across Stations and Directions

Figure C.1: Passenger Flows by Status during Morning Hours



(a) EBD Stations

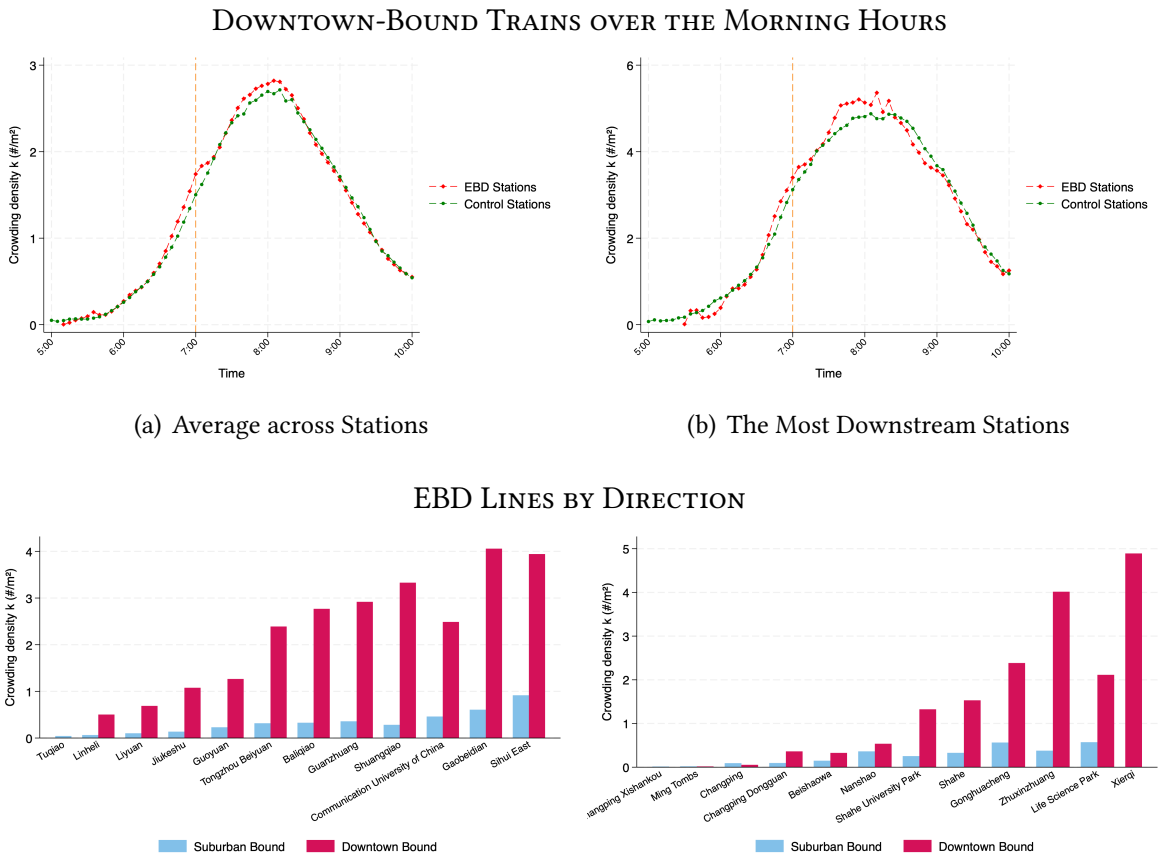
(b) Control Stations

Notes: This figure plots the number of passengers (in thousands) at EBD stations (panel a) and control stations (panel b) by 5-min time bin during morning hours. The four series represent passengers entering the station, exiting the station, transferring through the station, and riding in trains passing through the station. The orange dashed vertical line marks the 7 AM EBD cutoff, and the two gray dashed vertical lines mark 6:30 AM and 8:30 AM, which bound the sample period used in the analysis. Both groups of stations exhibit similar in-train ridership levels, peaking around 8:00 to 8:30 AM. At EBD stations, entry passengers display a visible dent just after 7 AM as some riders advance their departures to qualify for the discount, with a corresponding bunching just before 7 AM. No such discontinuity appears at control stations.

Because stations in our sample sit at the far ends of suburban lines, the vast majority of morning passengers travel toward downtown. The top row of Figure C.2 show the average in-train crowding density during morning hours in downtown-bound trains passing through EBD and control stations. The two groups exhibit strikingly similar levels and time profiles. Crowding density averages about 1.5 persons per square meter around 7 AM, peaks at roughly 2.8 persons per square meter near 8:15 AM, and declines to 0.5 persons per square meter by 10 AM. The EBD and control series nearly overlay except in a narrow window around the 7 AM cutoff, where bunching at EBD stations creates a visible hump. The right sub-panel narrows the comparison to the most downstream station on each line, where trains have accumulated passengers from all upstream stations. Although this sub-panel draws from only four stations (one from each line), the crowding patterns remain closely aligned between EBD and control lines. With passengers boarding from upstream stations, trains passing the most downstream stations are substantially more crowded. At around 8:20 AM, crowding density at the most downstream EBD stations peaks at approximately five persons per square meter.

The two graphs in the bottom row of Figure C.2 show that downtown-bound trains are far more

Figure C.2: In-Train Crowding Density



(a) Average across Stations

(b) The Most Downstream Stations

(c) Batong Line

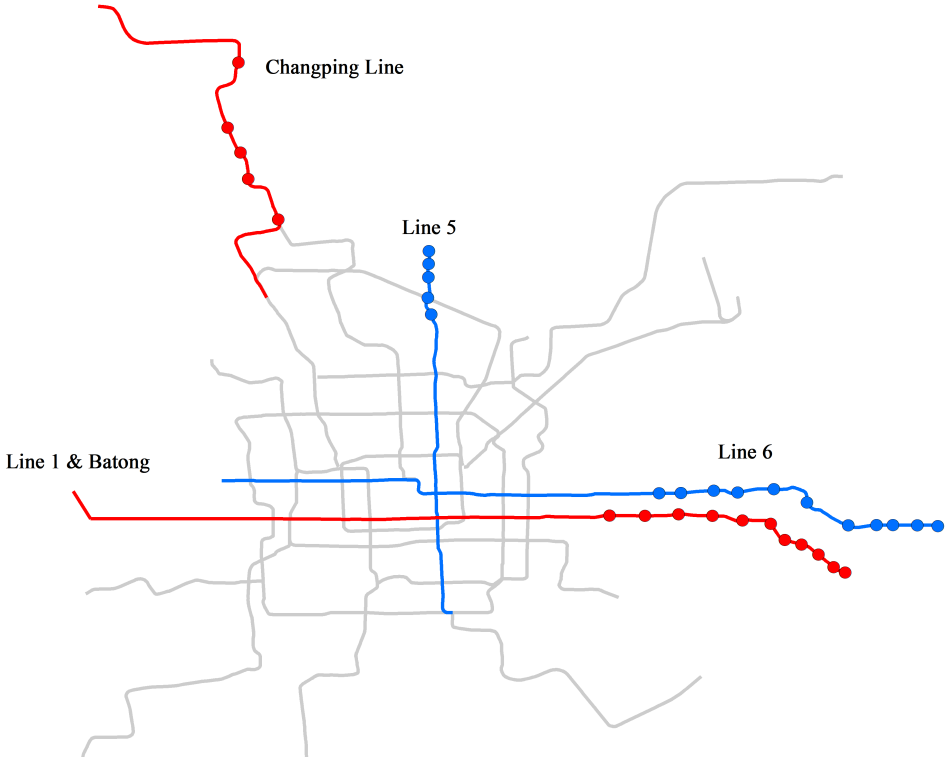
(d) Changping Line

Notes: This figure depicts in-train crowding density k (persons/m^2). The two graphs in the top row show downtown-bound trains over the course of the morning peak hours. Panel (a) averages across stations on EBD and control lines; panel (b) shows the most downstream station from each line. The two graphs in the bottom row show crowding density by station and direction for the two EBD lines during morning peak hours (6:30–8:30 AM). Stations are listed from the most upstream on the left to the most downstream on the right. Downtown-bound trains are substantially more crowded than suburban-bound trains. Crowding is computed using the full universe of rides.

crowded than suburban-bound trains on the two EBD lines. Trips in both directions qualify for the EBD and are included in our sample. The substantial difference in crowding between directions provides variation that helps identify potentially nonlinear crowding disutility at different density levels.

Passengers form expectations about crowding based on their commuting experience. To proxy for these expectations, we regress realized crowding on OD-by-day-of-week-by-time-bin fixed effects and use the fitted values $\widehat{\text{Crowd}}_{jt}$ as the expected crowding measure in the demand model. These fixed effects capture the systematic, predictable component of crowding patterns, precisely the information available to regular commuters. The same approach is applied to construct the expected travel time. The R^2 from this regression exceeds 0.98, indicating that passengers' crowding experiences are highly predictable from the OD pair, day of week, and time of day.

Figure C.3: Beijing Subway Map with EBD and Control Stations



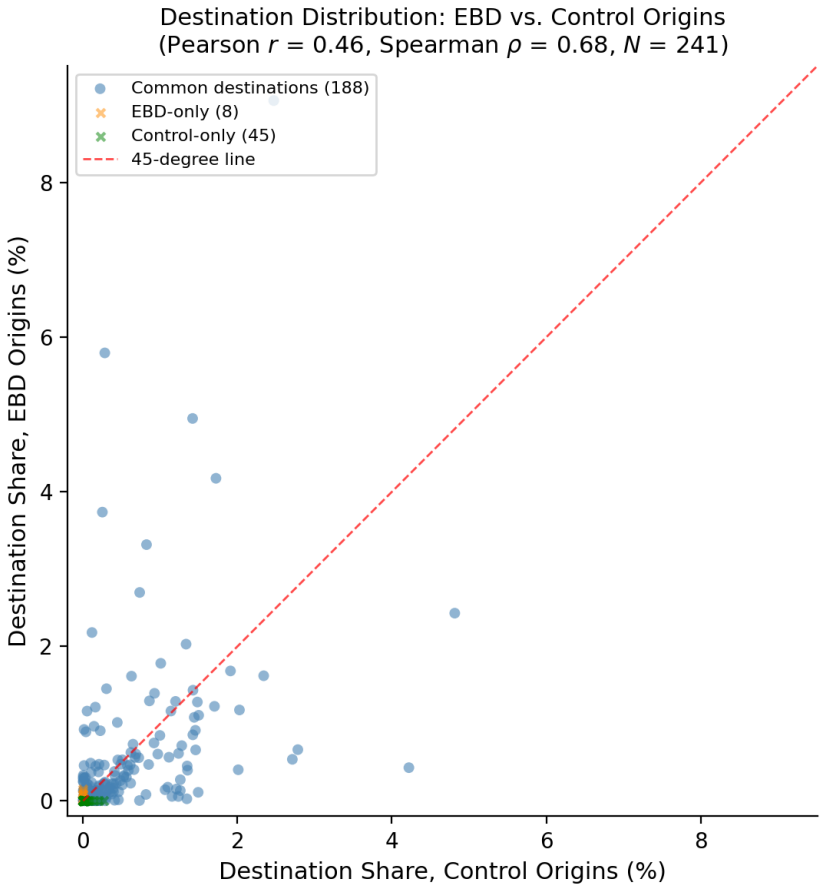
Notes: This figure shows the Beijing Subway network with the 16 early-bird discount (EBD) stations marked in red and the 16 control stations marked in blue. EBD stations are located on the Batong Line and the Changping Line. Control stations are on Line 5 and Line 6.

Table C.1: In-Station Time Estimation

	Estimated In-Station Time (min)				
	Mean	SD	P10	P90	N
Entry	3.50	1.84	1.36	5.34	33,436,600
Exit	2.00	0.90	0.89	3.15	33,436,600
Transfer	4.43	1.42	2.61	6.23	34,782,743

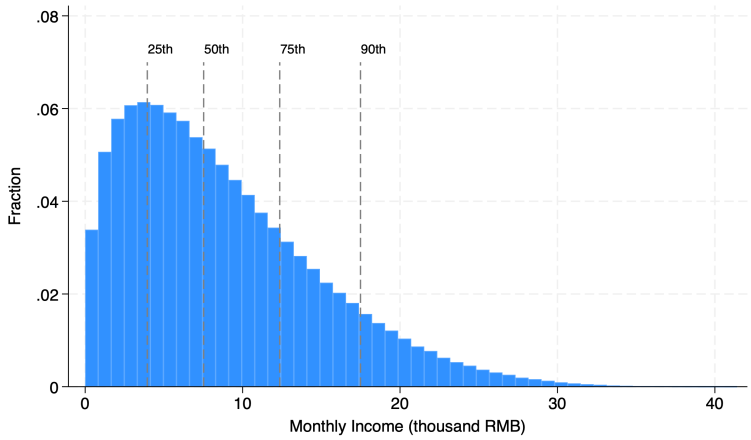
Notes: This table reports estimated in-station times from the iterative calibration procedure described in Section 4.1.1. Entry time is the duration from tapping in at the origin station to boarding the train. Exit time is the duration from alighting to tapping out, with the mean normalized to 2 minutes. Transfer time is the average duration across all transfer stations. N denotes the number of trips used in estimation.

Figure C.4: Destination Distribution of EBD and Control Station Riders



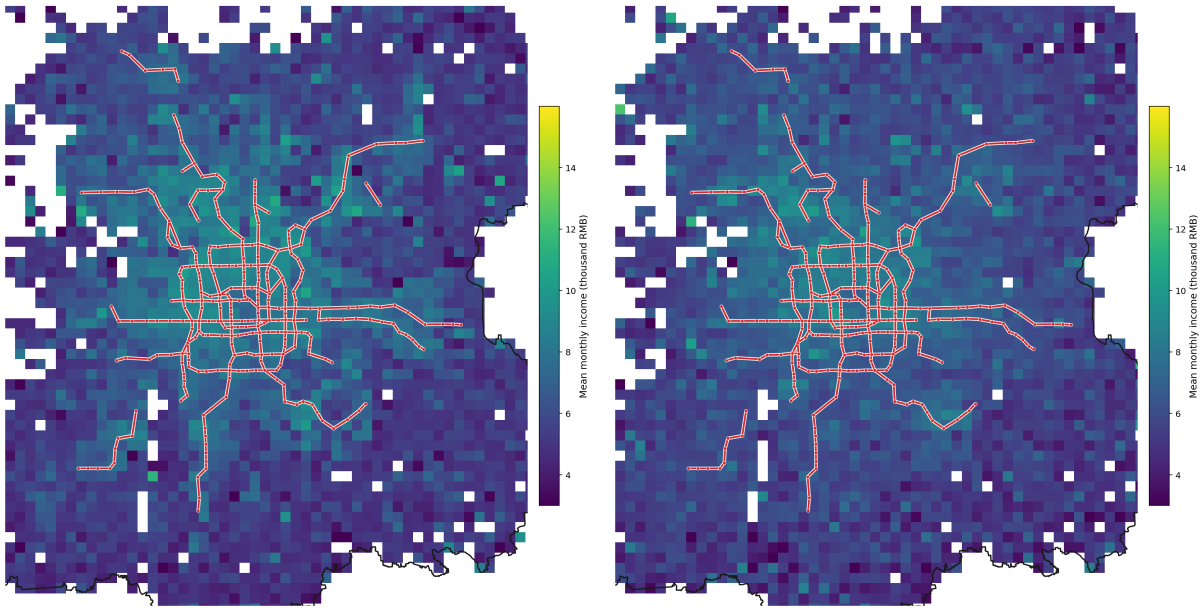
Notes: Each point represents a destination station. The horizontal axis is the share of all control-origin riders going to that destination; the vertical axis is the corresponding share for EBD-origin riders. Blue dots denote the 188 destinations served by both groups; orange and green crosses denote the 8 EBD-only and 45 control-only destinations, respectively. The red dashed line is the 45-degree line. Pearson $r = 0.46$, Spearman $\rho = 0.68$ ($N = 241$).

Figure C.5: Simulated Monthly Income of Subway Riders (Thousand RMB)



Notes: Distribution of simulated monthly incomes (in thousands of RMB) drawn from OD-specific beta distributions fitted to the Baidu commuting data. The weighted mean income is 8.77 thousand RMB.

Figure C.6: Mean Monthly Income by Cell

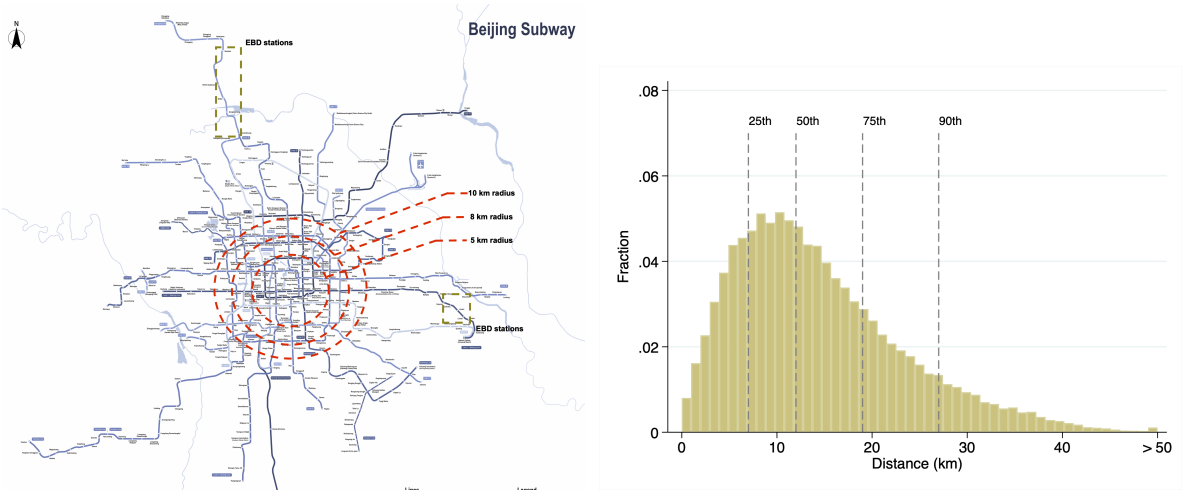


(a) Residential cell

(b) Workplace cell

Notes: Cell-level mean monthly income (thousand RMB, adjusted to 2016 levels) computed from the Baidu commuting portrait data as the population-weighted average across income brackets, with bracket midpoints as the per-bracket value. Cells are 2×2 aggregations of the underlying Baidu grid (cells of approximately 1.4 km on a side); cells with fewer than 5 commuters are dropped. The 2016 subway network (lines opened by 31 December 2016) is overlaid in red; the Beijing municipality boundary in dark grey.

Figure C.7: ICP Geography and Distance Distribution

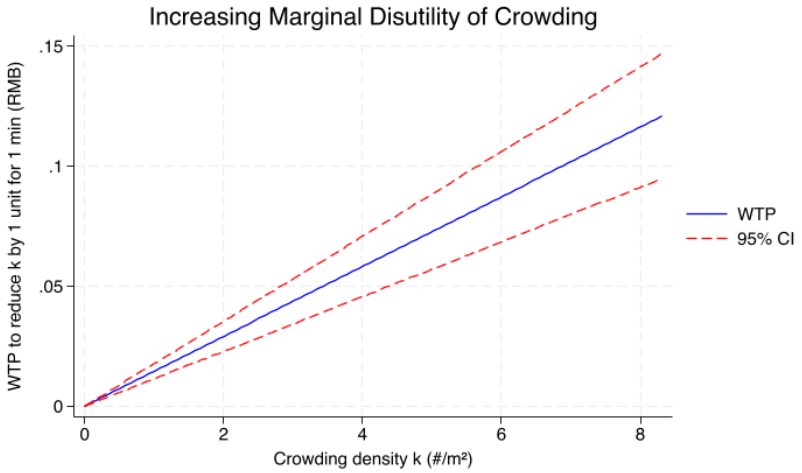


(a) Beijing Subway Network and Radius Rings

(b) Distance from ICP Destination D' to Focal Destination D (Infrequent Riders)

Notes: Panel (a) shows the Beijing Subway network. EBD stations, located on the Changping Line (north) and the Batong Line (southeast), are highlighted in dashed boxes at the far suburban ends of the network. Red dashed circles mark 5-km, 8-km, and 10-km radii from the city center. The vast majority of the network lies beyond the 8-km ring, illustrating that the exclusion zone around a focal destination D removes only a small fraction of potential ICPs. Panel (b) shows the distribution of distances between each infrequent ICP's destination D' and the focal passenger's destination D. Dashed vertical lines mark the 25th, 50th, 75th, and 90th percentiles. The median ICP destination is approximately 11 km from D, well beyond the 8-km exclusion threshold used in the preferred IV specification. Infrequent riders are defined as those taking ≤ 3 trips in the current week.

Figure C.8: Increasing Marginal Disutility of Crowding



Notes: The figure uses the overall-sample specification with OD-by-date and date-by-time-bin fixed effects in Table 3, column (6). Utility includes squared expected cumulative crowding exposure, instrumented by expected ICT and the four EBD policy interactions. The horizontal axis divides cumulative exposure by the sample mean in-train duration to express it as an equivalent density. The solid line plots marginal WTP per person-minute per square meter; dashed lines show the 95% confidence interval.

Table C.2: Constant Coefficient Model Estimates: Frequent Riders

	OLS: Uninstrumented	GMM: ICT only		GMM: ICT + EBD instruments		GMM: Nonlinear
		1st Stage	2nd Stage	1st Stage	2nd Stage	2nd Stage
	(1)	(2)	(3)	(4)	(5)	(6)
Crowding	0.0062		−0.0201		−0.0094	
	(0.0003)		(0.0019)		(0.0011)	
Crowding^2						−0.000028
						(0.000003)
Price	−0.2093	−5.1519	−0.3280	29.3990	−0.1821	−0.1513
	(0.0151)	(0.4331)	(0.0196)	(2.1966)	(0.0094)	(0.0090)
Rescheduling	−0.0299	0.0354	−0.0274	−0.1360	−0.0250	−0.0252
	(0.0010)	(0.0306)	(0.0014)	(0.0243)	(0.0010)	(0.0010)
ICT		11.5338		11.1407		
		(0.4788)		(0.4645)		
Other IVs				✓		
Wald F-Test			580.2		284.8	441.8
R^2	0.6588		−0.3272		−0.0720	
OD-Date FE	X	X	X	X	X	X
Date-Time FE	X	X	X	X	X	X
Mean Rescheduling (mins)	4.654		4.654		4.654	
Mean Crowding (person·min/m^2)	127.1		127.1		127.1	
Implied MWTP						
Crowding	−0.0296		0.0613		0.0516	*see*
	(0.00266)		(0.00550)		(0.00613)	*Fig. C.8*
Rescheduling	0.143		0.0835		0.137	0.1662
	(0.0114)		(0.00689)		(0.00894)	(0.0120)

Notes: This table mirrors Table 3 in the main text but restricts the estimation sample to frequent riders, defined as passengers who take at least four subway trips in the current week. All specifications include OD-by-date and date-by-time-bin fixed effects. Column (1) reports uninstrumented OLS estimates. In columns (2)–(3), crowding is instrumented by ICT from infrequent riders with $|D - D'| \geq 8$ km. In columns (4)–(6), four EBD policy interactions are added as instruments (Treat $\times 1\{t < 7{:}00\}$, Treat $\times$ trend, Treat $\times$ trend^2, Treat $\times$ trend^3). Column (6) includes a quadratic crowding term. Standard errors in parentheses, clustered at the OD level. MWTP is computed as the ratio of the relevant coefficient to the price coefficient. The estimates are within roughly ten percent of the overall-sample baseline reported in Table 3, confirming that the constant-coefficient results are stable across these two sample definitions. Crowding and ICT denote expected values; the row labels omit "expected".

Table C.3: Constant Coefficient Model Estimates Using ICP

	GMM: ICP only		GMM: ICP + EBD instruments	
	1st Stage (1)	2nd Stage (2)	1st Stage (3)	2nd Stage (4)
Crowding		−0.0175		−0.0083
		(0.0019)		(0.0011)
Price	−4.9237	−0.3075		−0.1730
	(0.4180)	(0.0185)		(0.0087)
Rescheduling	0.0305	−0.0279	−0.1365	−0.0252
	(0.0279)	(0.0012)	(0.0218)	(0.0009)
ICP	178.1106		179.4798	
	(8.0566)		(7.8686)	
Other IVs			✓	
Wald *F*-Test		488.7		302.9
Observations	321,496	321,496	321,496	321,496
OD-Date FE	X	X	X	X
Date-Time FE	X	X	X	X
Implied Willingness to Pay				
MWTP for Crowding Reduction		0.0569		0.0482
		(0.0054)		(0.0060)
MWTP for Resched. Avoidance		0.0907		0.146
		(0.0071)		(0.0090)

Notes: This table reports GMM estimates using the number of incidental companion passengers (ICP) rather than incidental companion time (ICT) as the instrumental variable for crowding. Columns (1) and (3) report the first-stage regressions of in-train crowding on ICP under the ICP-only and ICP-plus-EBD specifications, respectively; columns (2) and (4) report the corresponding second-stage demand estimates with ln(share) as the dependent variable. All columns include OD-by-date and date-by-time-bin fixed effects. The sample is the overall sample of riders. Standard errors in parentheses, clustered at the OD level. MWTP computed as the ratio of the relevant coefficient to the price coefficient. Crowding and ICP denote expected values; the row labels omit "expected".

Table C.4: Robustness: In-Station Crowding

	ICT only In-Station as Control (1)	ICT + EBD instruments In-Station as Endogenous (2)
In-train crowding	−0.0293	−0.0078
	(0.0026)	(0.0013)
Price	−0.5855	−0.1618
	(0.0323)	(0.0097)
Rescheduling	−0.0291	−0.0242
	(0.0014)	(0.0011)
In-station crowding	0.0403	−0.0077
	(0.0030)	(0.0023)
Mean in-train crowding (person·min/m^2)	127.09	127.09
Mean in-station crowding (thousand person·min)	14.31	14.31
Observations	295,400	295,400
Wald F-stat	424.40	470.58
MWTP in-train crowding (RMB per person·min/m^2)	0.0500	0.0482
	(0.0029)	(0.0075)
MWTP in-station crowding (RMB per thousand person·min)	−0.0688	0.0476
OD-Date FE	X	X
Date-Time FE	X	X

Notes: Dependent variable is $\ln(S_{jdt})$ from the linear estimating equation (4), estimated by GMM. Standard errors clustered at the OD level. The in-station crowding regressor InStn_{jdt} is the product-level mean of passengers encountered during entry, transfer, and exit, weighted by the duration of overlap, expressed in thousands of person-minutes. Column (1) adds InStn_{jdt} as an uninstrumented control under the ICT-only instrument set; this specification is mis-specified because InStn_{jdt} is itself endogenous, and the positive coefficient on InStn_{jdt} reflects that endogeneity. Column (2) treats InStn_{jdt} as a second endogenous regressor, jointly instrumented with in-train crowding by the ICT-plus-EBD set: ICT (the infrequent-companion-time instrument, restricted to companions whose destinations lie at least 8 km from the focal D) plus the four EBD policy interactions treat $\times\, 1\{t < 7\text{am}\}$ and treat $\times\, t^k$ for $k = 1, 2, 3$. The sample consists of frequent riders. The two "Mean" rows report regression-sample averages of cumulative in-train crowding exposure (person·min/m^2) and total in-station ridership-time (thousand person·min). WTP for in-train crowding reduction is $\hat{\beta}_{\text{crowd}}/\hat{\alpha}$, reported with delta-method standard errors. Crowding and ICT denote expected values; the row labels omit "expected".

Table C.5: First Stage Estimation for Crowding (ICT + EBD Instruments)

	(1)	(2)	(3)
ICT	10.9955	10.9076	10.5954
	(0.4318)	(0.4441)	(0.4185)
Treat × Pre-7am	51.7127	51.7809	35.3100
	(3.4115)	(3.4145)	(3.4639)
Treat × Trend	534.2271	533.2125	590.7363
	(34.8764)	(34.8727)	(38.4703)
Treat × Trend^2	-17.6784	-17.6444	-19.3772
	(1.1699)	(1.1699)	(1.2872)
Treat × Trend^3	0.1930	0.1926	0.2096
	(0.0130)	(0.0130)	(0.0143)
Price	29.1758	29.2149	19.1122
	(2.0850)	(2.0870)	(2.0233)
Rescheduling	-0.1331	-0.1363	-0.1343
	(0.0219)	(0.0221)	(0.0190)
Observations	321,496	321,496	321,496
OD-Date FE	X	X	X
Time FE	X		
Date-Time FE		X	
Time-Destin Line FE			X

Notes: This table reports the first stage of the ICT-plus-EBD GMM specification for the overall sample of riders. The dependent variable is expected in-train crowding (person·min/m^2). Instruments are ICT from infrequent riders with $|D - D'| \geq 8$ km, plus four EBD policy interactions. Standard errors in parentheses, clustered at the OD level. Columns (1)–(3) differ only in the time fixed effects: time-bin (Column 1), date-by-time-bin (Column 2, used as the preferred specification), and time-bin-by-destination-line (Column 3). Crowding and ICT denote expected values; the row labels omit "expected".

C.2 Crowding and Travel Time

Column (1) of Table C.6 reports a regression of individual trip travel time on the average in-train crowding density (measured in persons per square meter) and the total number of in-station passengers (measured in counts), controlling for OD-by-date and line-time fixed effects. Each observation is an individual trip, and there are some 33 million observations. Increasing in-train crowding density by one person per square meter, an increase of roughly 35% from the mean, raises travel time by only 0.27 minutes. Given an average travel time of about 39 minutes, this effect represents a mere 0.7% increase. The regression includes the number of passengers the focal rider encountered during the trip, ensuring that the coefficient on crowding density captures the relationship between *in-train* crowding and travel time, which is the margin relevant for the demand model, rather than confounding it with delays caused by congestion in station corridors, platforms, and gates.

To put the magnitude in perspective, we convert the delay cost into monetary terms. The median monthly earnings for full-time workers is approximately 8,000 RMB. Assuming 160 work hours per month, this implies an earning rate of about 0.8 RMB per minute. Using the rule-of-thumb value of time in transit to be 50% of the wage rate, or 0.4 RMB per minute, the 0.27-minute increase in travel time from a one-person-per-square-meter increase in crowding translates to a monetary cost of about 0.11 RMB. As we show in the estimation results, this is an order of magnitude smaller than the willingness to pay for reducing discomfort from crowding.

The relationship between in-train crowding and delay is slightly convex. Column (2) of Table C.6 reports results from a quadratic specification. The linear term is negative (-0.16) and the quadratic term is positive (0.05), indicating that crowding has essentially no effect on travel time at low density levels and a small positive effect at high density levels. Figure C.9 plots the implied total monetary cost of crowding-induced delay as a function of crowding density. Below the 90th percentile of crowding density, the delay cost remains below 0.3 RMB, a small fraction of the average fare and the WTP for crowding reduction estimated later.

In-station crowding is also positively associated with total travel time, though it is not the focus of this paper. A one-person increase in in-station crowding is associated with a 0.003 minute increase in travel time. To compare the magnitudes, note that an increase of one person per square meter in in-train crowding density represents about a 35% increase from the ridership-weighted sample average of 2.87 persons/m^2. A comparable 35% increase from the mean in-station crowding of 668 persons corresponds to an increase of 234 persons. This translates to an increase in total travel time of $234 \times 0.0034 \approx 0.80$ minutes, or about 0.32 RMB in value of time. This is still small, but larger than that from in-train crowding. The fact that an increase in in-station crowding produces a larger delay than the same proportional increase in in-train crowding suggests that crowding delays travel primarily through slower walking traffic inside stations rather than by slowing trains down. This is intuitive, as pedestrian flows through corridors, stairways, and platform gates are directly impeded

Table C.6: The Effects of Crowding on Travel Time

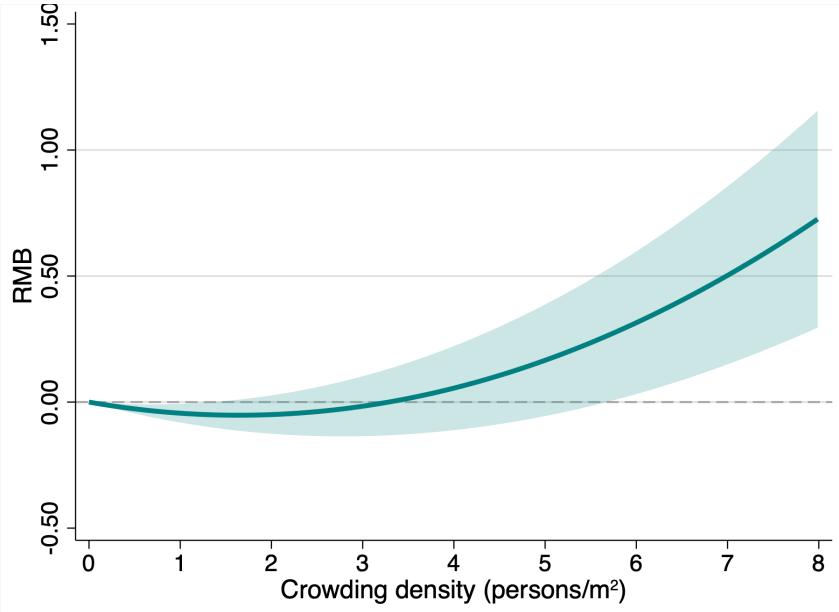
	Travel Time	
	Linear (1)	Quadratic (2)
Crowding density k (persons/m^2)	0.2704	−0.1586
	(0.0176)	(0.0468)
Crowding density k squared		0.0483
		(0.0063)
In-station crowding (persons)	0.0034	0.0037
	(0.0000)	(0.0001)
N	33,436,600	30,280,427
Mean dep. var.	38.59	38.05
Mean crowding density k (persons/m^2)	2.87	2.87
Mean in-station crowding (persons)	667.72	667.72
OD-Date FE	X	X
Line-Time FE	X	X

Notes: Regressions of trip travel time on in-train crowding density (persons/m^2) and in-station crowding (number of passengers). Column (1) is the linear specification on the full sample. Column (2) is the quadratic specification, restricted to cells with at least two trips. Standard errors in parentheses, clustered at the OD level.

by larger crowds, whereas train speed is unaffected by passenger load.

While this paper focuses on in-train crowding, we note that in-station crowding also causes discomfort and generates disutility. In-station and in-train crowding are positively correlated, as a passenger experiencing a crowded platform can generally expect a crowded train. Although we do not directly estimate the disutility from in-station crowding, we show in the robustness checks that in-station crowding does not confound the estimated WTP to avoid crowded trains.

Figure C.9: Total Monetary Cost of Crowding-Induced Delay

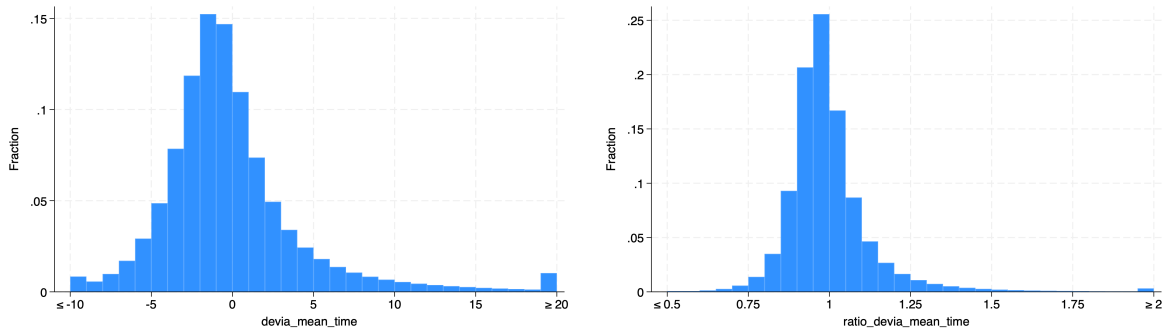


Notes: This figure plots the implied delay cost (in RMB) from the quadratic specification in Column (2) of Table C.6 as a function of crowding density. Value of time is assumed to be 50% of median earnings (0.4 RMB/min). Delay cost remains below 0.3 RMB for crowding levels up to the 90th percentile.

C.3 Evidence on Rerouting

First, Figure C.10 shows the distribution of trip-level deviations from the OD-specific mean travel time. Panel (a) reports the deviation in minutes; panel (b) reports the ratio of individual travel time to the OD mean. The distributions are tightly concentrated around zero. The vast majority of trips cluster close to the mean, with 65.1% of trips falling within 3 minutes, 83.6% within 5 minutes, and 95.8% within 10 minutes. When expressed as a ratio to the mean, 71.6% of trips fall in the interval [0.9, 1.1] and 94.8% in [0.75, 1.25]. If a substantial fraction of riders were taking alternative routes, we would expect to see a cluster of travelers in the far right tail, as rerouting causes considerable delay. No such cluster is apparent.

Figure C.10: Histograms of Deviation from the Mean Travel Time



(a) Absolute Deviation (Minutes)

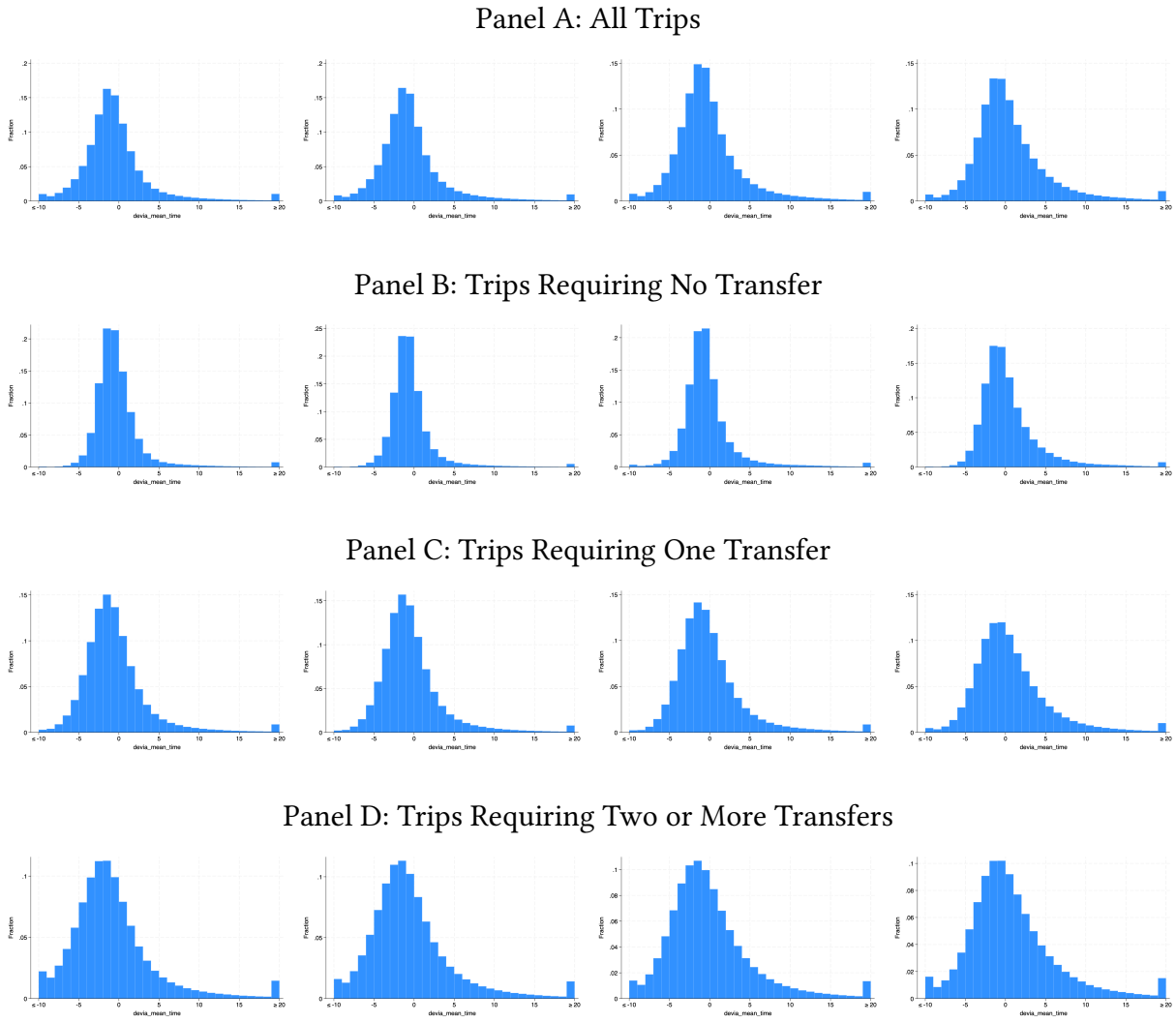
(b) Ratio to Mean Travel Time

Notes: Panel (a) shows the absolute deviation of each trip's travel time from the OD-specific mean (in minutes). Panel (b) shows the ratio of trip travel time to the OD-specific mean. Sample includes all trips in the working sample.

Alternative routes typically cost considerably more time than the shortest path, so if crowding systematically causes rerouting, it should have a large effect on the dispersion of travel time. Figure C.11 shows the distribution of travel time as a function of crowding. As in-train crowding increases, the distribution of trip time becomes slightly wider, indicating a larger variance, but the increase is not visually salient and there is no cluster of riders in the right tail at higher crowding quartiles. Much of the widening is driven by crowded stations slowing down foot traffic rather than by larger crowds slowing down trains. For trips that do not require a transfer, the distribution is much tighter and crowding does not lead to visually discernible increases in the variance. Trips requiring one or more transfers have larger variances in travel time to begin with, and crowding produces somewhat more visible increases in spread. Crucially, even for trips with transfers, the distributions remain similar across crowding quartiles, indicating that the right tail grows with the *number of transfers* rather than with *crowding levels*.

Second, we show formally in regressions that crowding has a negligible impact on the travel time variance. Table C.7 tests this directly. The dependent variables are the squared deviation (column

Figure C.11: Distribution of Travel Time Deviations by Crowding Quartile and Number of Transfers



Notes: Each row shows the distribution of deviations from the OD-specific mean travel time (in minutes) for a given transfer category, split by quartiles of in-train crowding density (least crowded on the left, most crowded on the right). Panel A includes all trips; Panel B includes trips without transfers; Panel C includes trips with one transfer; Panel D includes trips with two or more transfers. The distributions remain similar across crowding quartiles within each transfer category, indicating that the right tail grows with the number of transfers rather than with crowding levels.

1), absolute deviation (column 2), and ratio to mean (column 3) of each trip's travel time from its OD-date-time-bin cell mean. All regressions control for OD-by-date and line-time fixed effects and include both in-train crowding density and in-station ridership.

The coefficients on crowding density are statistically significant but economically small. In column 1, a one-person per square meter increase in crowding density raises the squared deviation of travel time from its cell mean by 0.60 minutes^2, about 3.7% of the sample mean of 16.38 minutes^2. The same one-unit increase is associated with a 0.12-minute increase in absolute deviation (column 2), relative to a mean absolute deviation of 2.40 minutes. In column 3, the correlation between crowding

Table C.7: The Effects of Crowding on Travel Time Variance

	(1) Squared Deviation	(2) Absolute Deviation	(3) Ratio to Mean
Crowding density k (persons/m^2)	0.6014	0.1244	-0.0000
	(0.1289)	(0.0090)	(0.0000)
In-station ridership (persons)	0.0023	0.0004	0.0000
	(0.0003)	(0.0000)	(0.0000)
N	30,280,427	30,280,427	30,280,427
Mean dep. var.	16.38	2.40	1.00
Mean crowding density k (persons/m^2)	2.87	2.87	2.87
Mean in-station ridership (persons)	667.72	667.72	667.72
OD-Date FE	X	X	X
Line-Time FE	X	X	X

Notes: Each observation is a trip. The dependent variable in column (1) is the squared deviation of the trip's travel time from its cell mean, $(TT_i - \bar{TT}_{\text{cell}})^2$, where a cell is defined as an OD-date-15-minute-bin. Column (2) uses the absolute deviation $|TT_i - \bar{TT}_{\text{cell}}|$. Column (3) uses the ratio of the trip's travel time to its cell mean, $TT_i/\bar{TT}_{\text{cell}}$. Cells with fewer than 2 trips are excluded, yielding 30,280,427 observations. Standard errors in parentheses, clustered at the OD level.

and ratio to the mean is essentially zero. These patterns are inconsistent with systematic rerouting, which would generate large right-tail outliers and substantial increases in travel time dispersion.

D Details of Welfare and Counterfactual Analysis

This appendix collects the numerical implementation details for the counterfactual exercises of Section 7 and reports sensitivity analyses with respect to the car allocation between Business and Standard classes.

D.1 Numerical Implementation

Our counterfactual computations rely on three numerical primitives: one-dimensional bisection for root finding, damped fixed-point iteration for the two-class ridership vector, and a scan-bracket-bisect procedure to enforce revenue neutrality. We summarize each here.

One-dimensional roots are computed by bisection. Given a continuous function F and an interval $[a, b]$ with $F(a)F(b) < 0$, we iterate $m = (a+b)/2$ and retain the sub-interval that preserves the sign change until $|F(m)|$ falls below a tolerance set between 10^{-9} and 10^{-12} depending on the application. We use this routine in three places: to invert the private demand mapping, to locate the Pigouvian social optimum, and to compute the queuing equilibrium wait.

The inverse private demand $PD(N)$ is evaluated as follows. For each candidate N, we bisect on P to solve $f(P; N) = \sum_g M_g\, s_g(A + \alpha_g P + \beta_g LN) - N = 0$. The function $f(\cdot; N)$ is continuous in P and strictly decreasing when $\alpha_g < 0$, which guarantees a unique root in a sufficiently wide bracket. The baseline calibration of A uses the same routine in reverse: we bisect on A so that the implicit solution of $PD(N) = P_{\text{base}}$ equals N_{eq}.

The two-class ridership vector (N_B, N_S) at prices (P_B, P_S) is obtained by damped fixed-point iteration. Starting from a feasible initial guess, we update

$$N_B^{(k+1)} \;\leftarrow\; (1-\eta)N_B^{(k)} + \eta \sum_g M_g\, p_{gB}\big(V_{gB}(N_B^{(k)}), V_{gS}(N_S^{(k)})\big),$$

and analogously for $N_S^{(k+1)}$, with damping parameter $\eta \in [0.5, 0.7]$. Damping keeps the iterations stable when the nested-logit probabilities have steep local slopes in the class shares. Convergence is declared when both components change by less than 10^{-8} between successive iterations.

Revenue neutrality for the two-class counterfactual requires simultaneous control over P_B, P_S, and the equilibrium ridership vector. For each candidate P_S, we scan P_B upward from P_S to bracket every sign change in $R(P_B, P_S) - R_{\text{base}}$, where $R_{\text{base}} = P_{\text{base}} \times N_{\text{eq}} = 6{,}533$ RMB is the revenue target. Within each bracket we bisect to find a revenue-neutral P_B, and among all revenue-neutral pairs (P_B, P_S) we retain the one that maximizes the variety-adjusted consumer surplus. A final one-dimensional refinement over P_S yields the reported optimum.

Sensitivity to λ is reported in Table D.1. As λ rises, the variety bonus grows, and CS_{VA} falls toward baseline consumer surplus. At $\lambda = 0.8$, CS_{VA} is only +0.77 percent above baseline because revenue

neutrality forces $P_B \approx P_S$: the nested logit with a high λ penalizes unequal prices too strongly for self-selection to be sustained. At $\lambda = 0.3$, the gain reaches 15.6 percent, though such low nesting parameters are at the extreme end of the plausible range. The preferred value $\lambda = 0.5$ yields a 7.82 percent variety-adjusted welfare gain and is a conservative choice within the typical range.

Table D.1: Two-Class Pricing: Sensitivity to the Nesting Parameter λ

λ	P_B	P_S	N	VB	CS_{VA}	$\Delta\text{CS}_{\text{VA}}$ (%)
0.3	10.73	2.51	1538	1434	12579	15.56
0.4	10.63	2.48	1532	1918	12133	11.47
0.5	10.30	2.44	1524	2405	11737	7.82
0.6	9.74	2.43	1514	2896	11403	4.76
0.7	8.95	2.46	1505	3390	11144	2.38
0.8	7.76	2.64	1495	3888	10968	0.77

Notes: All scenarios maintain revenue neutrality (R = 6,533 RMB) with a 3+3 Business/Standard car allocation. VB denotes the variety bonus at the revenue-neutral equal price, $CS_{\text{VA}} = CS_{\text{total}} - \text{VB}$ is the variety-adjusted consumer surplus, and ΔCS_{VA} is the percentage change relative to baseline consumer surplus (10,885 RMB).

D.2 Sensitivity to Car Allocation

We examine how the two-class results vary with the Business/Standard split, holding $\lambda = 0.5$ fixed and maintaining revenue neutrality. A clear threshold emerges from Table D.2. When the number of Standard cars exceeds the number of Business cars, as in the 1+5 and 2+4 splits, the welfare optimizer fails to generate meaningful price differentiation. The revenue-neutral welfare-maximizing prices collapse to $P_B \approx P_S$, the price differential is numerically indistinguishable from zero, and the variety-adjusted consumer surplus gain vanishes. The mechanism is direct. When the Standard class has excess capacity, per-rider crowding in Standard is already low at the revenue-neutral equal price, so riders have little incentive to pay a premium for Business class. The optimizer cannot sustain self-selection because the crowding differential between classes disappears mechanically.

Splits with $J_B \geq J_S$ produce substantial price differentiation. Under the current calibration, the preferred 3+3 allocation yields $P_B^\star = 10.30$, $P_S^\star = 2.44$, with a variety-adjusted consumer surplus gain of 7.82 percent (Table D.1, row $\lambda = 0.5$). Other splits in Table D.2 (1+5, 2+4, 4+2, 5+1) have not been re-run under the current calibration; the qualitative threshold remains: revenue-neutral two-class pricing is effective only when $J_B \geq J_S$. Below this threshold, revenue neutrality forces the two-class system to collapse to a single uniform fare.

Table D.2: Two-Class Pricing: Sensitivity to Car Allocation

	$J_S > J_B$			$J_B \geq J_S$	
	1+5	2+4	3+3	4+2	5+1
$P_B^\star$ (yuan)	4.46	4.43	6.98	5.77	4.80
$P_S^\star$ (yuan)	4.43	4.43	2.97	2.96	3.57
$P_B^\star - P_S^\star$ (yuan)	0.03	0.00	4.01	2.81	1.22
Total ridership (N)	1,506	1,508	1,543	1,560	1,523
Price discrim. gain (RMB)	−2.7	−0.6	108.1	231.1	58.0
$\Delta \text{CS}_{\text{VA}}$ (%)	−0.03	−0.01	+1.17	+2.50	+0.63

Notes: All scenarios maintain revenue neutrality (R = 6,533 RMB) with λ = 0.5 and $J_B + J_S = 6$. Price discrimination gain is the difference between CS at the welfare-maximizing revenue-neutral prices and CS at the revenue-neutral equal price. ΔCS_{VA} is the percentage change relative to baseline single-class consumer surplus (10,885 RMB; the value shown for the 3+3 row reflects the current calibration, while the other rows are pre-recalibration and have not been regenerated).